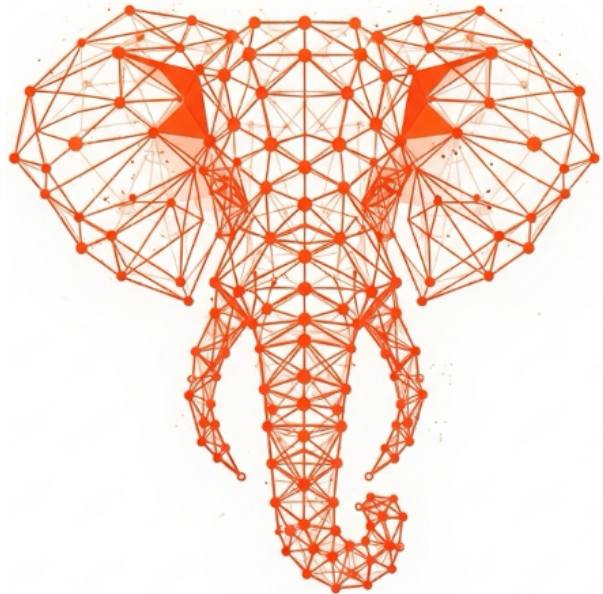




Uni
ct **DEEP LEARNING**
ADVANCED MODELS AND METHODS

Deep Learning

Advanced Models & Methods

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Corso di Laurea Magistrale in Informatica

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Università di Catania

Self-Supervised Learning

Problem: Supervised Learning is Expensive!

Assume you want to label 1M images. How much will it cost?

(1,000,000 images)

× (10 seconds/image)

× (1/3600 hours/second)

× (\$15 / hour)

= \$41,667

(Small to medium sized dataset)

(Fast annotation)

(Low wage paid to annotator)

(Other assumptions: one annotator per image, no benefits / payroll tax / crowdsourcing fee for annotators; not accounting for time to set up tasks for annotators, etc. Real costs could easily be 3x this or more)

Problem: Supervised Learning is Expensive!

Assume you want to label **1B** images. How much will it cost?

(1,000,000,000 images)

(Large dataset)

× (10 seconds/image)

(Fast annotation)

× (1/3600 hours/second)

× (\$15 / hour)

(Low wage paid to annotator)

= **\$41,666,667**

(Other assumptions: one annotator per image, no benefits / payroll tax / crowdsourcing fee for annotators; not accounting for time to set up tasks for annotators, etc. Real costs could easily be 3x this or more)

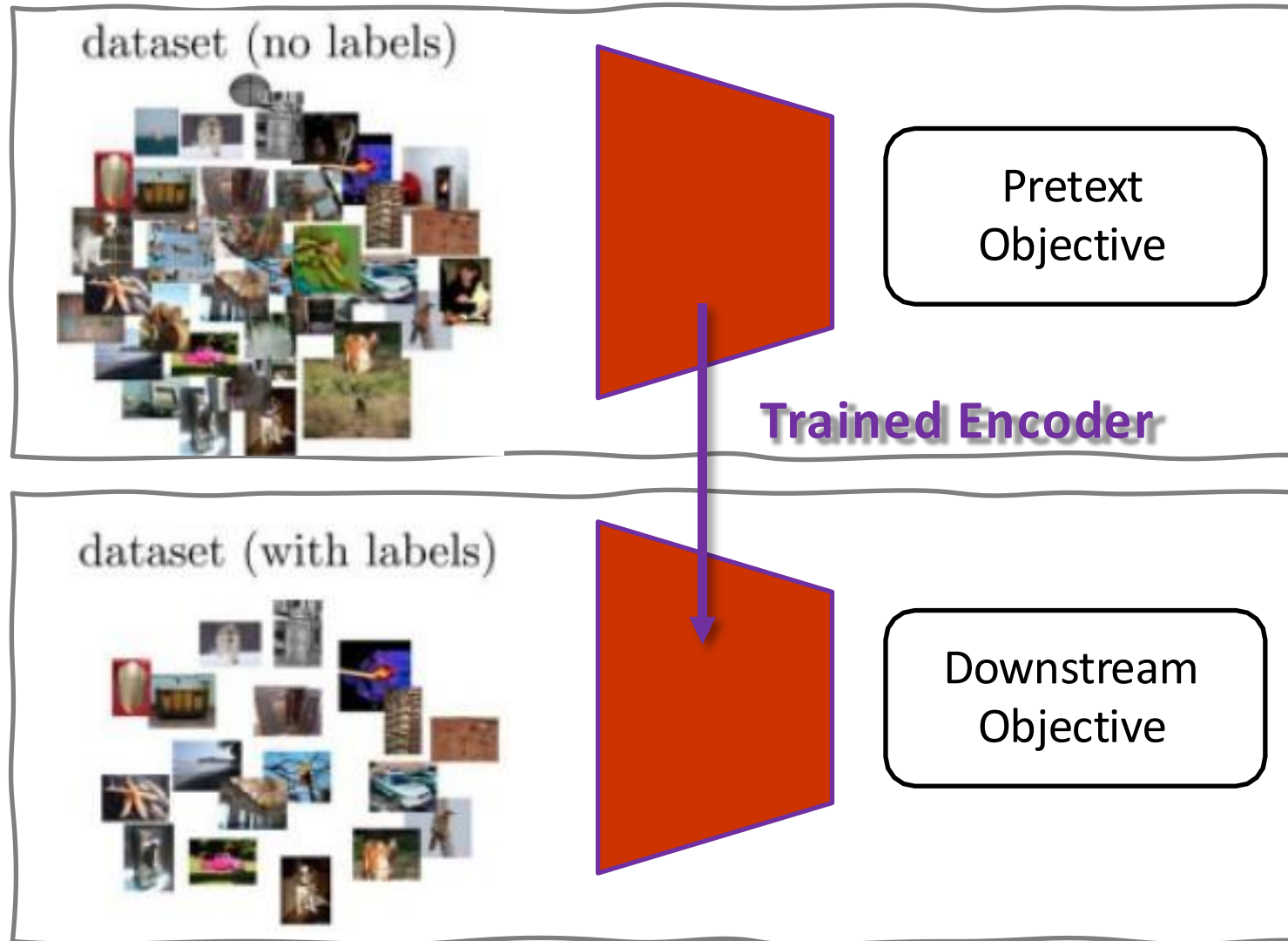
Problem: Supervised Learning is Not How We Learn

Babies don't get supervision for everything they see!



[Baby image](#) is [CC0 public domain](#)

Self-Supervised Learning



Pretext Task

- Define a task based on the data itself
- No manual annotation
- Could be considered an **unsupervised** task;
- but we learn with supervised learning objectives, e.g., classification or regression.

Downstream Task

- The application you care about
- You do not have large datasets
- The dataset is labeled
- Often linear probing

Self-supervised pretext tasks

A pretext task is a supervised task that uses automatically generated labels from raw data to learn useful representations — without requiring manual annotation.

Example: learn to predict image transformations / complete corrupted images

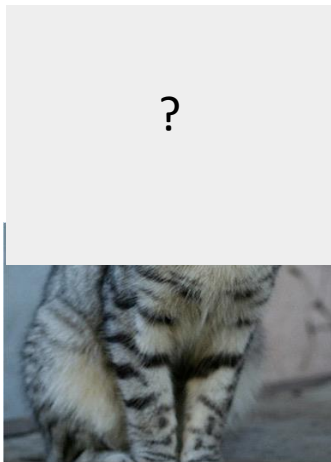
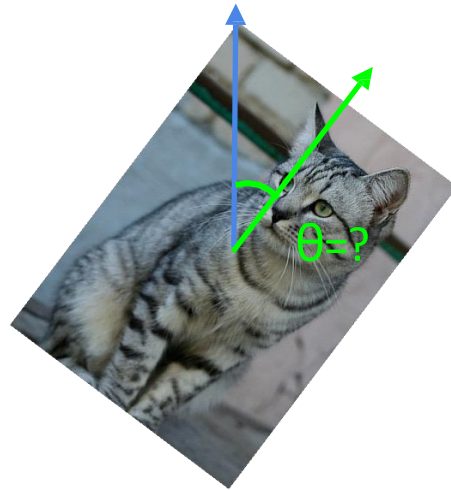


image completion



rotation prediction



jigsaw puzzle

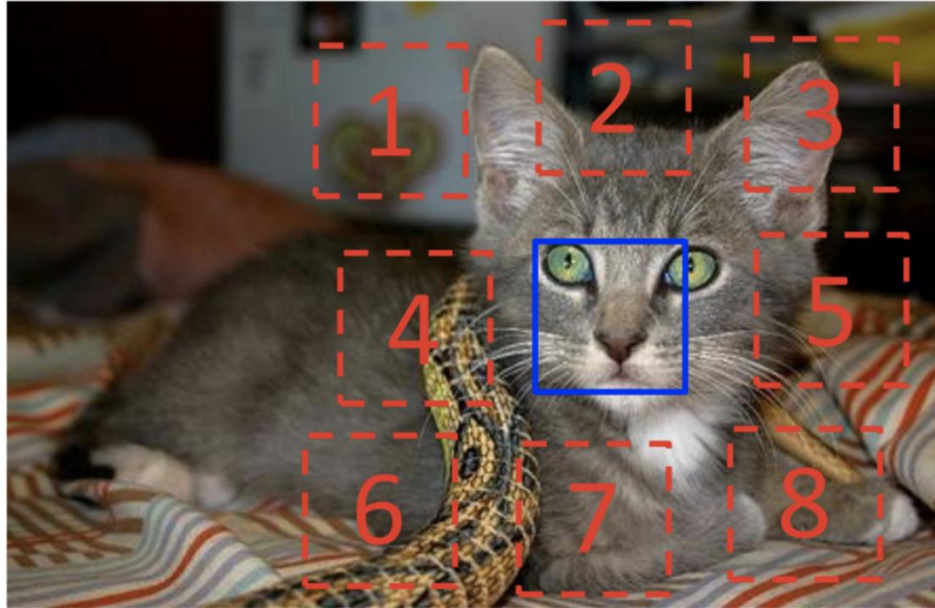


colorization

1. Solving the pretext tasks allows the model to learn good features.
2. We can automatically generate labels for the pretext tasks.

Pre-text Tasks

Pretext task: predict relative patch locations



$$X = (\text{patch 4}, \text{patch 5}); Y = 3$$

Example:



Question 1:



Question 2:



(Image source: [Doersch et al., 2015](https://arxiv.org/abs/1506.02136))

Classification over 8 positions

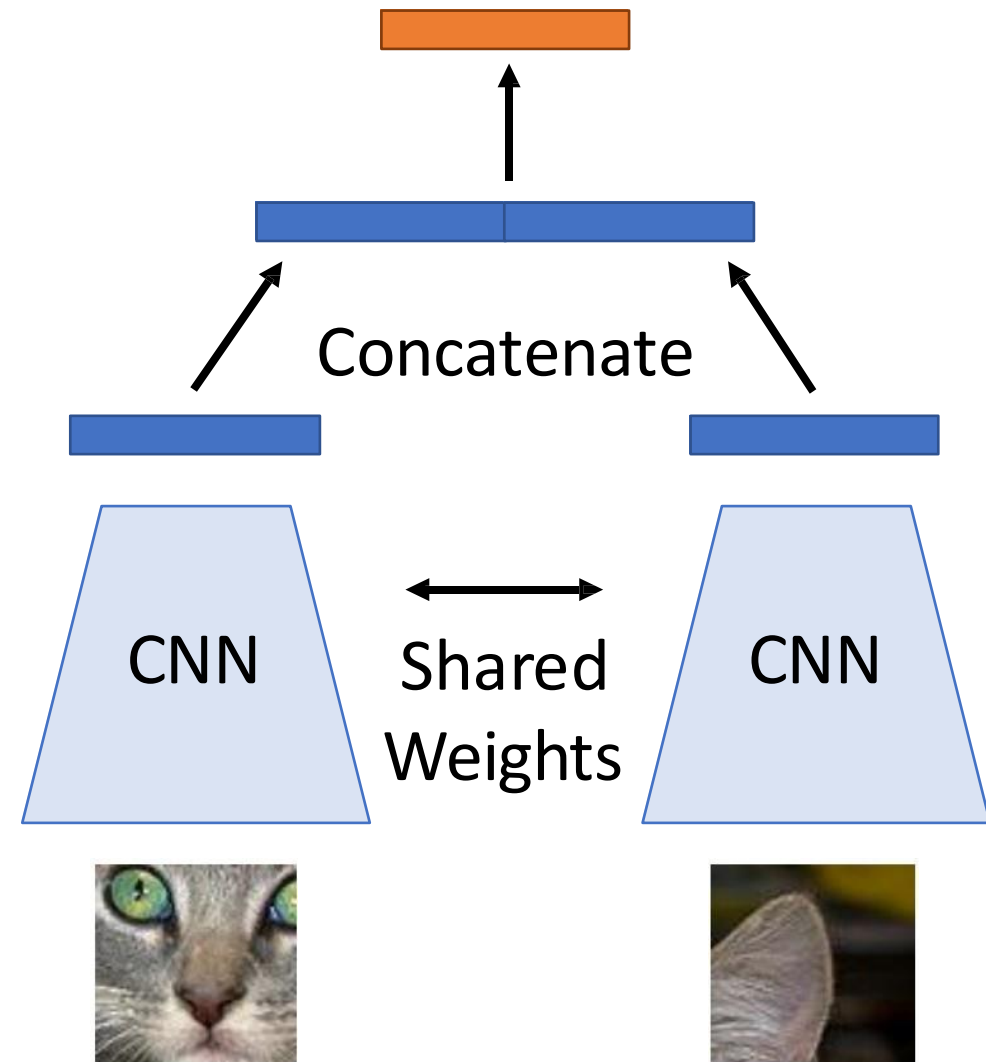
Context Prediction

Model predicts relative location of two patches from the same image.

Discriminative pretraining task

Intuition: Requires understanding objects and their parts

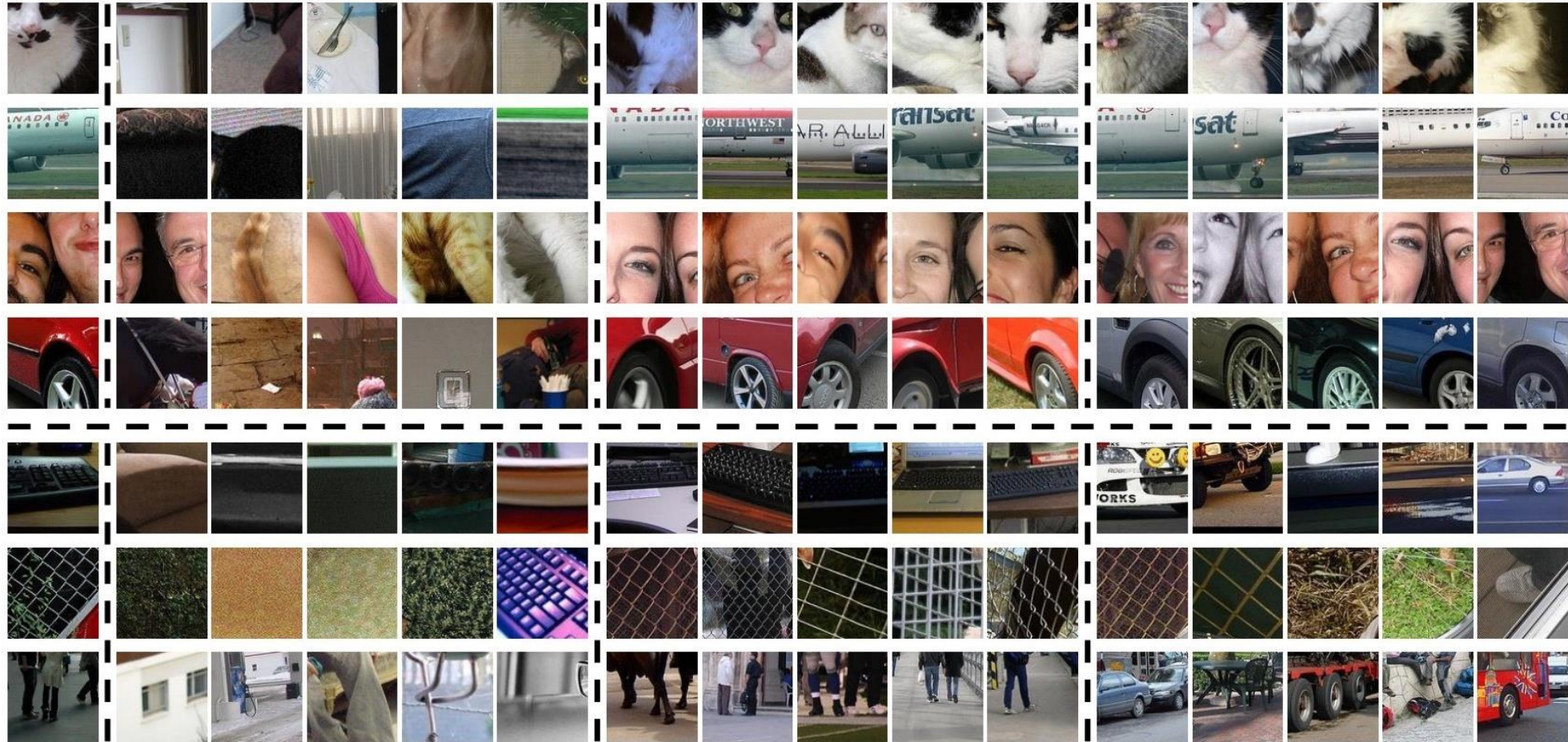
Two networks with shared weights is often called a "Siamese network"



Doersch et al, "Unsupervised Visual Representation Learning by Context Prediction", ICCV 2015

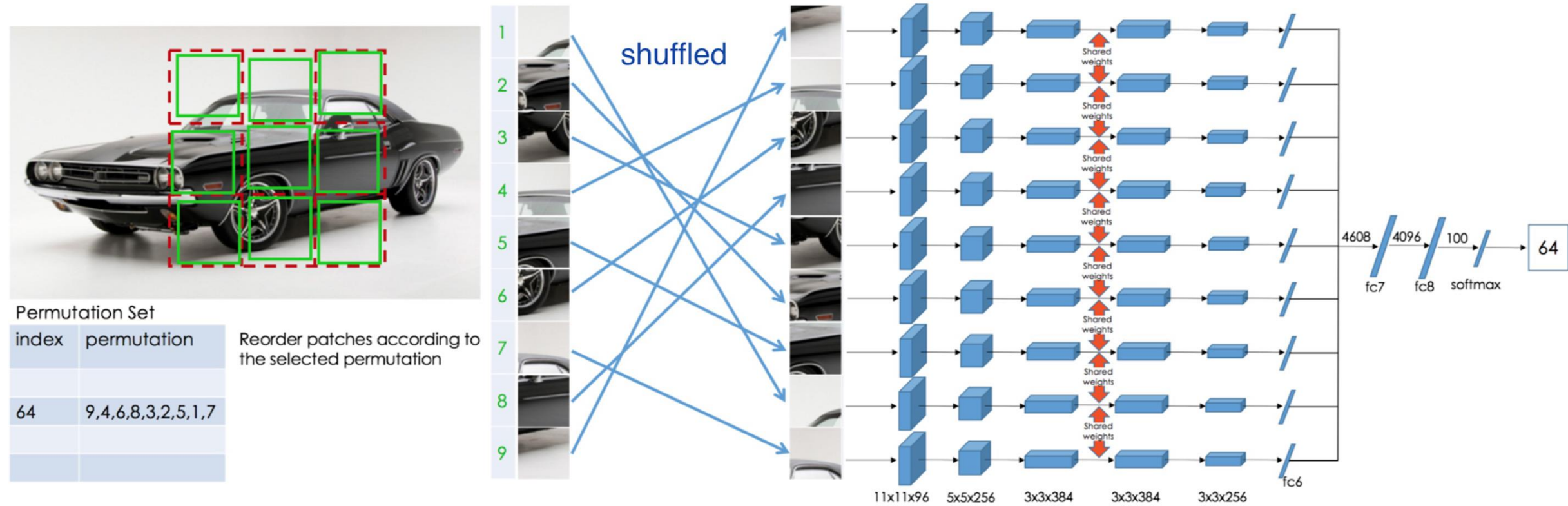
Context Prediction: Nearest Neighbors in Feature Space

Input Patch Random Init Supervised AlexNet Their Features



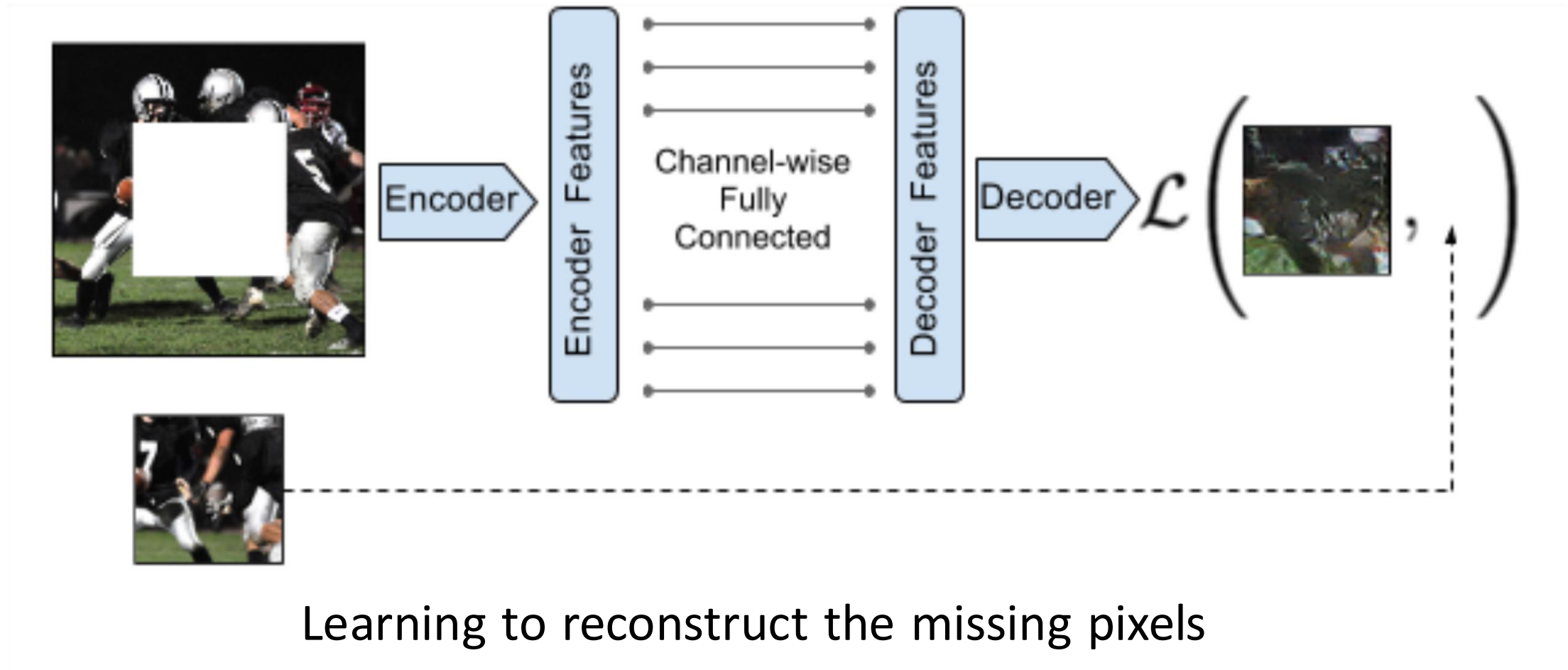
Doersch et al, "Unsupervised Visual Representation Learning by Context Prediction", ICCV 2015

Pretext task: solving “jigsaw puzzles”



(Image source: [Noroozi & Favaro, 2016](#))

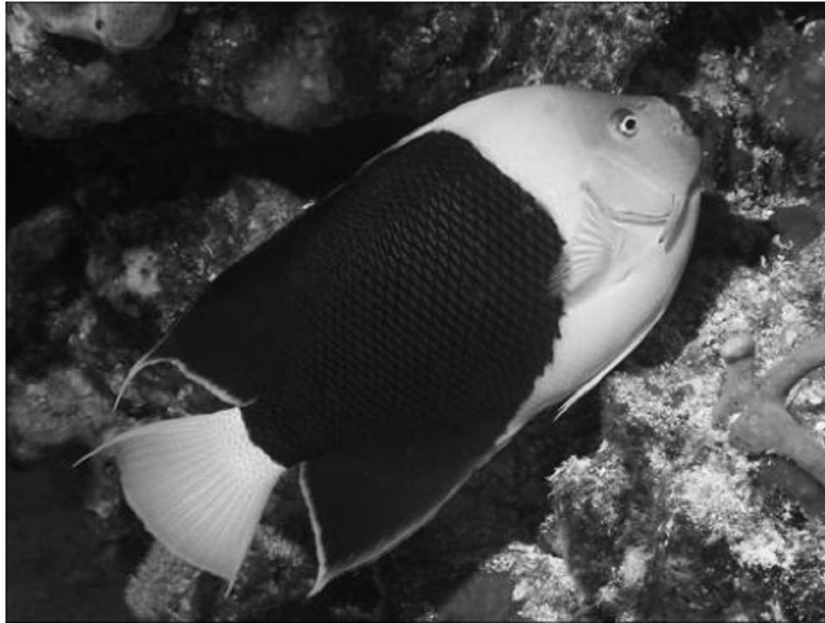
Learning to inpaint by reconstruction



Learning to reconstruct the missing pixels

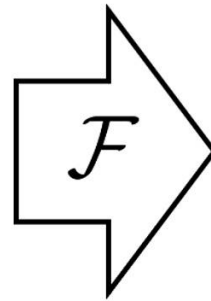
Source: [Pathak et al., 2016](#)

Pretext task: image coloring



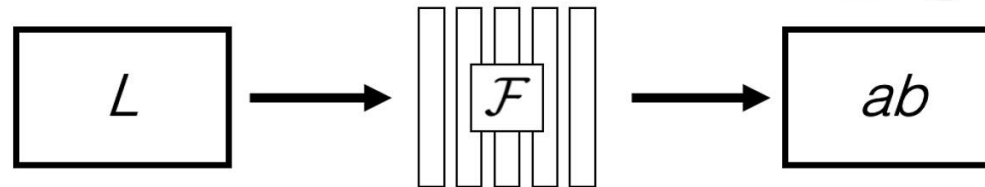
Grayscale image: L channel

$$\mathbf{X} \in \mathbb{R}^{H \times W \times 1}$$



Color information: ab channels

$$\hat{\mathbf{Y}} \in \mathbb{R}^{H \times W \times 2}$$



Source: Richard Zhang / Phillip Isola

Pretext task: image coloring



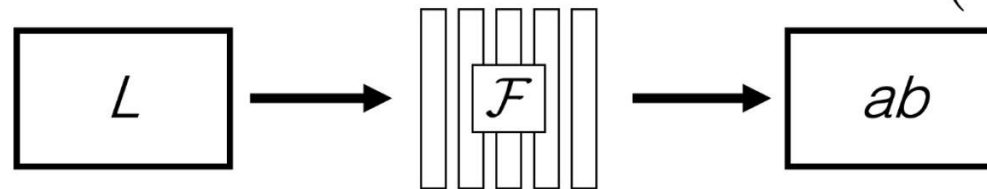
Grayscale image: L channel

$$\mathbf{X} \in \mathbb{R}^{H \times W \times 1}$$



Concatenate (L, ab) channels

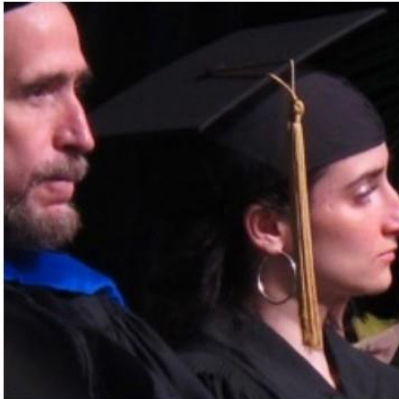
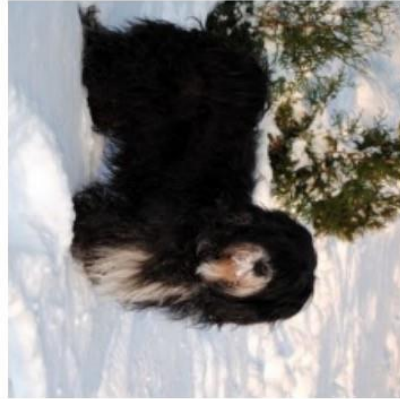
$$(\mathbf{X}, \hat{\mathbf{Y}})$$



Source: Richard Zhang / Phillip Isola

RotNet: Predict Rotation

4-way classification task: How much was each image rotated? (0, 90, 180, or 270 degrees)



90

270

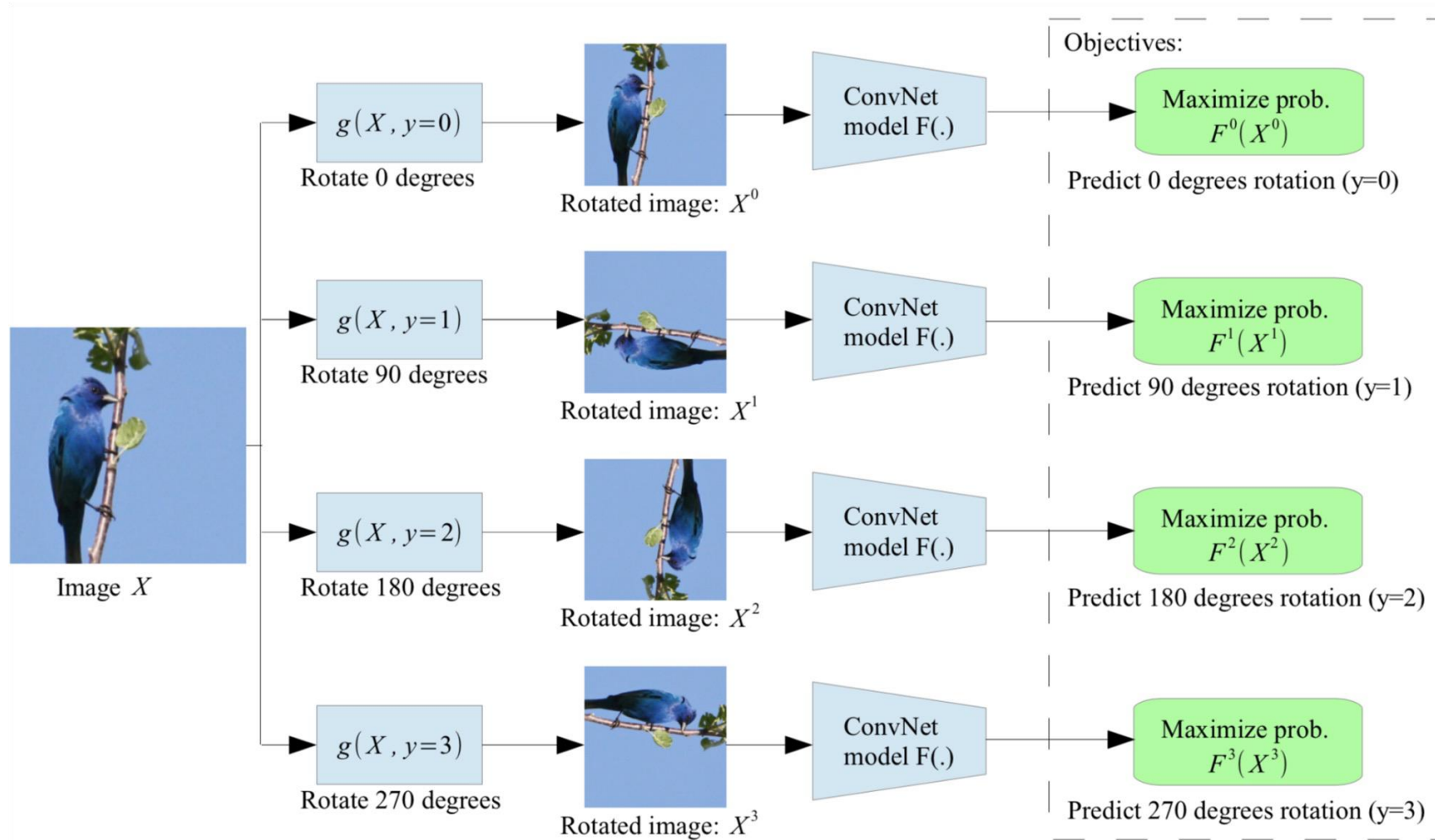
180

0

270

Gidaris et al, "Unsupervised representation learning by predicting image rotations", ICLR 2018

Pretext task: predict rotations



Self-supervised learning by rotating the entire input images.

The model learns to predict which rotation is applied (4-way classification)

(Image source: [Gidaris et al. 2018](#))

Transfer learned features to supervised learning

	Classification (%mAP)		Detection (%mAP)	Segmentation (%mIoU)
Trained layers	fc6-8	all	all	all
ImageNet labels	78.9	79.9	56.8	48.0
Random		53.3	43.4	19.8
Random rescaled Krähenbühl et al. (2015)	39.2	56.6	45.6	32.6
Egomotion (Agrawal et al., 2015)	31.0	54.2	43.9	
Context Encoders (Pathak et al., 2016b)	34.6	56.5	44.5	29.7
Tracking (Wang & Gupta, 2015)	55.6	63.1	47.4	
Context (Doersch et al., 2015)	55.1	65.3	51.1	
Colorization (Zhang et al., 2016a)	61.5	65.6	46.9	35.6
BIGAN (Donahue et al., 2016)	52.3	60.1	46.9	34.9
Jigsaw Puzzles (Noroozi & Favaro, 2016)	-	67.6	53.2	37.6
NAT (Bojanowski & Joulin, 2017)	56.7	65.3	49.4	
Split-Brain (Zhang et al., 2016b)	63.0	67.1	46.7	36.0
ColorProxy (Larsson et al., 2017)		65.9		38.4
Counting (Noroozi et al., 2017)	-	67.7	51.4	36.6
(Ours) RotNet	70.87	72.97	54.4	39.1

Pretrained with full ImageNet supervision

No pretraining

Self-supervised learning on ImageNet (entire training set) with AlexNet.

Finetune on labeled data from Pascal VOC 2007.

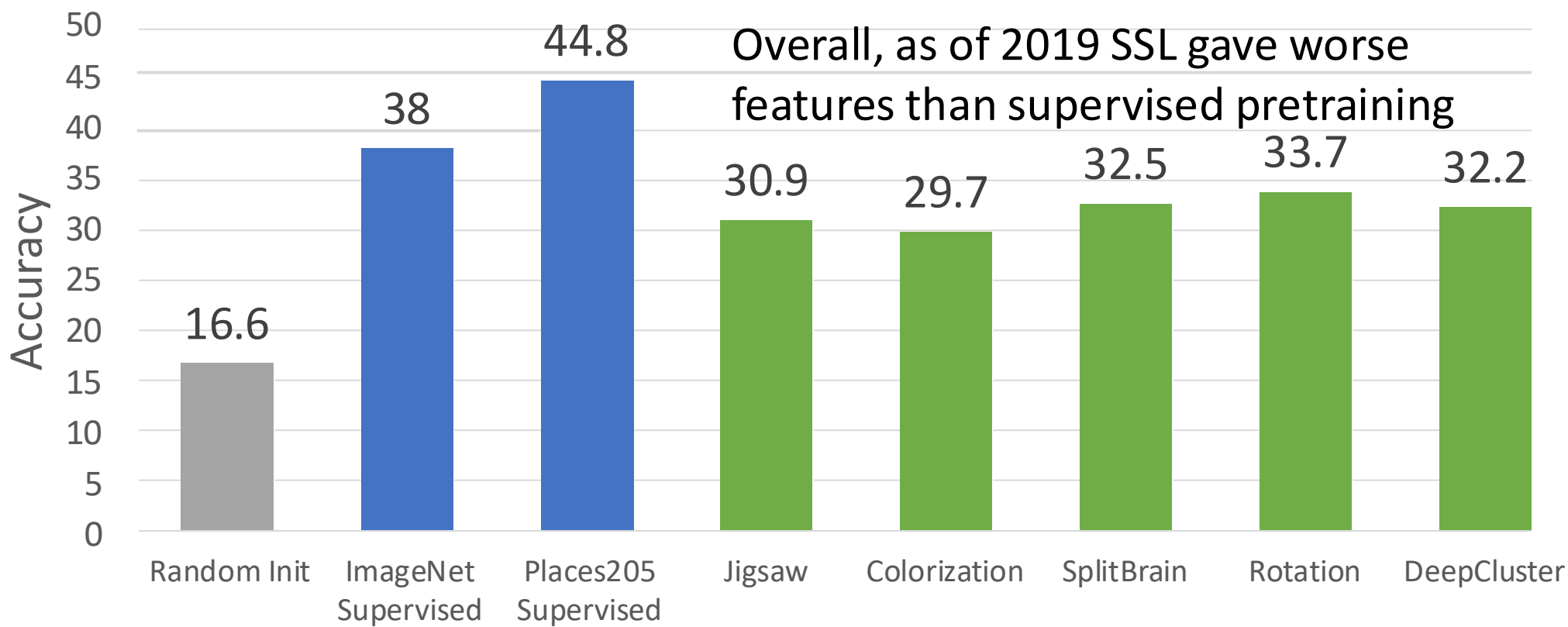
Self-supervised learning with rotation prediction

source: [Gidaris et al. 2018](#)

Which SSL Method is best?

Some papers have tried to perform fair comparisons of many SSL methods

Places205 Linear Classification from AlexNet conv5



Goyal et al, "Scaling and Benchmarking Self-Supervised Visual Representation Learning", ICCV 2019

Summary: pretext tasks from transformations

- Pretext tasks focus on “visual common sense”, e.g., predict rotations, inpainting, rearrangement, and colorization.
- The models are forced learn good features about natural images, e.g., semantic representation of an object category or region correspondences in order to solve the pretext tasks.
- We often do not care about the performance of these pretext tasks, but rather how useful the learned features are for downstream tasks (classification, detection, segmentation).
- **Problems:**
 - 1) coming up with individual pretext tasks is tedious
 - 2) the learned representations may not be general.

Pretext tasks from image transformations

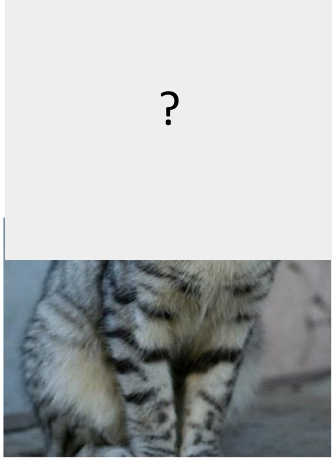
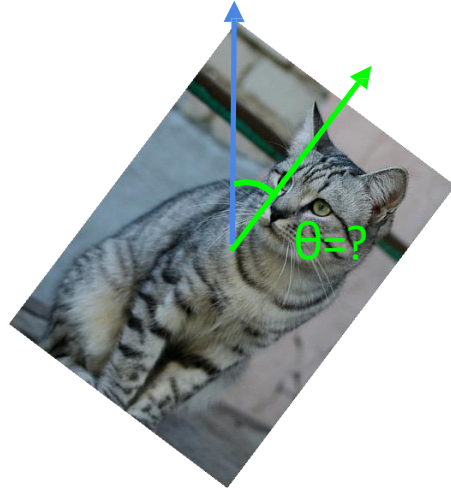


image completion



rotation prediction



“jigsaw puzzle”



colorization

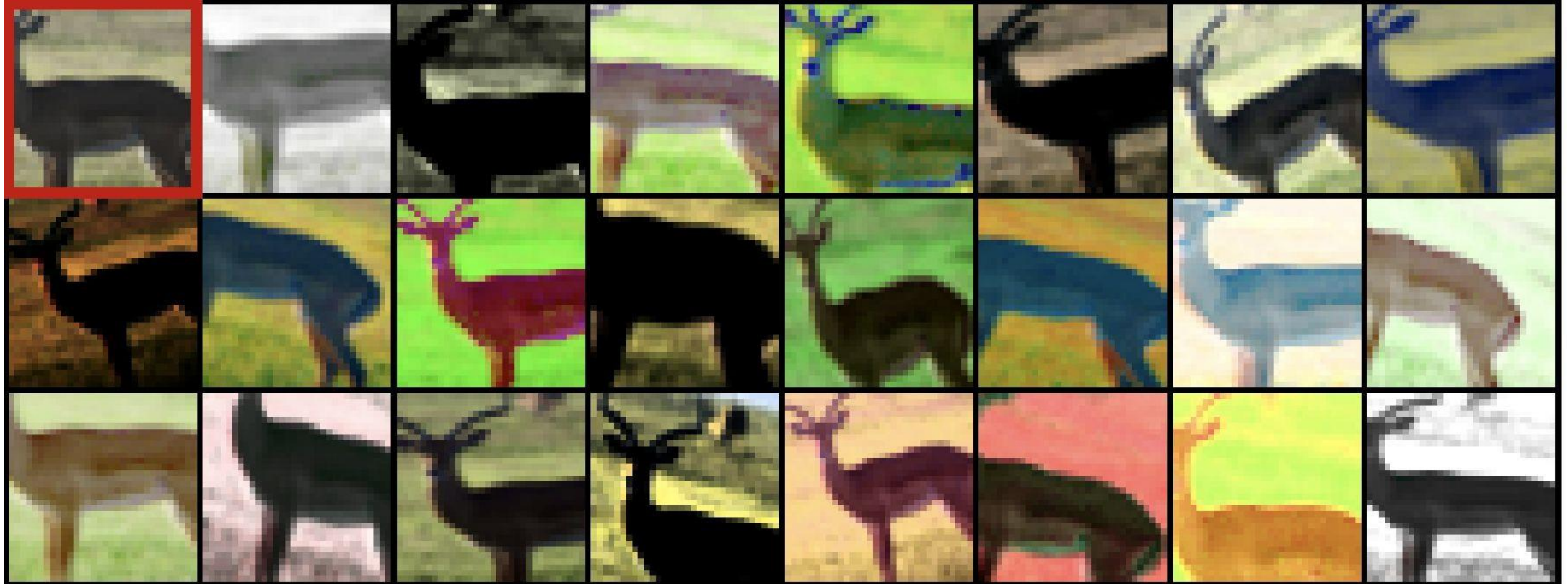
Learned representations may be tied to a specific pretext task!

Can we come up with a more general pretext task?

Contrastive Approaches

Invariance-Based Learning

Learn representations that are invariant to data transformations



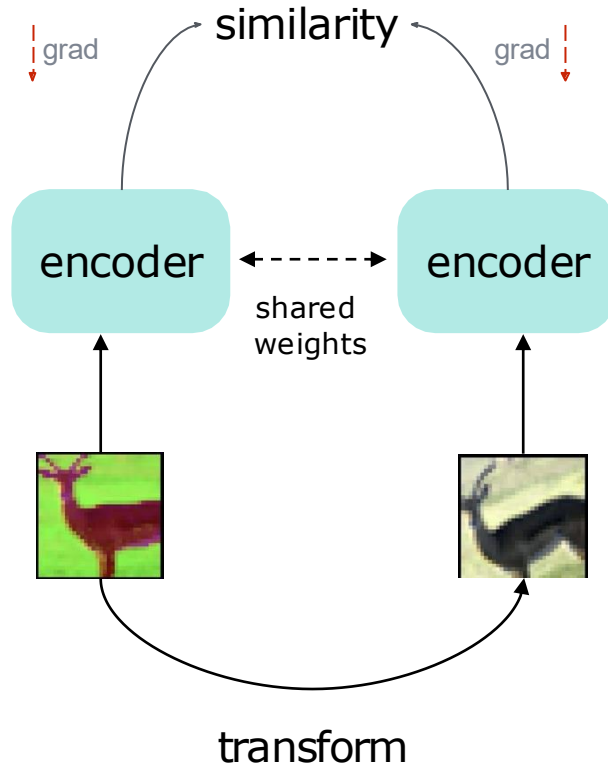
Learn features such that:

$$f_{\theta}(I) = f_{\theta}(\text{augment}(I))$$

Figure from Dosovitskiy et al., 2014

Simple Approach

$$f_{\theta}(I) = f_{\theta}(\text{augment}(I))$$



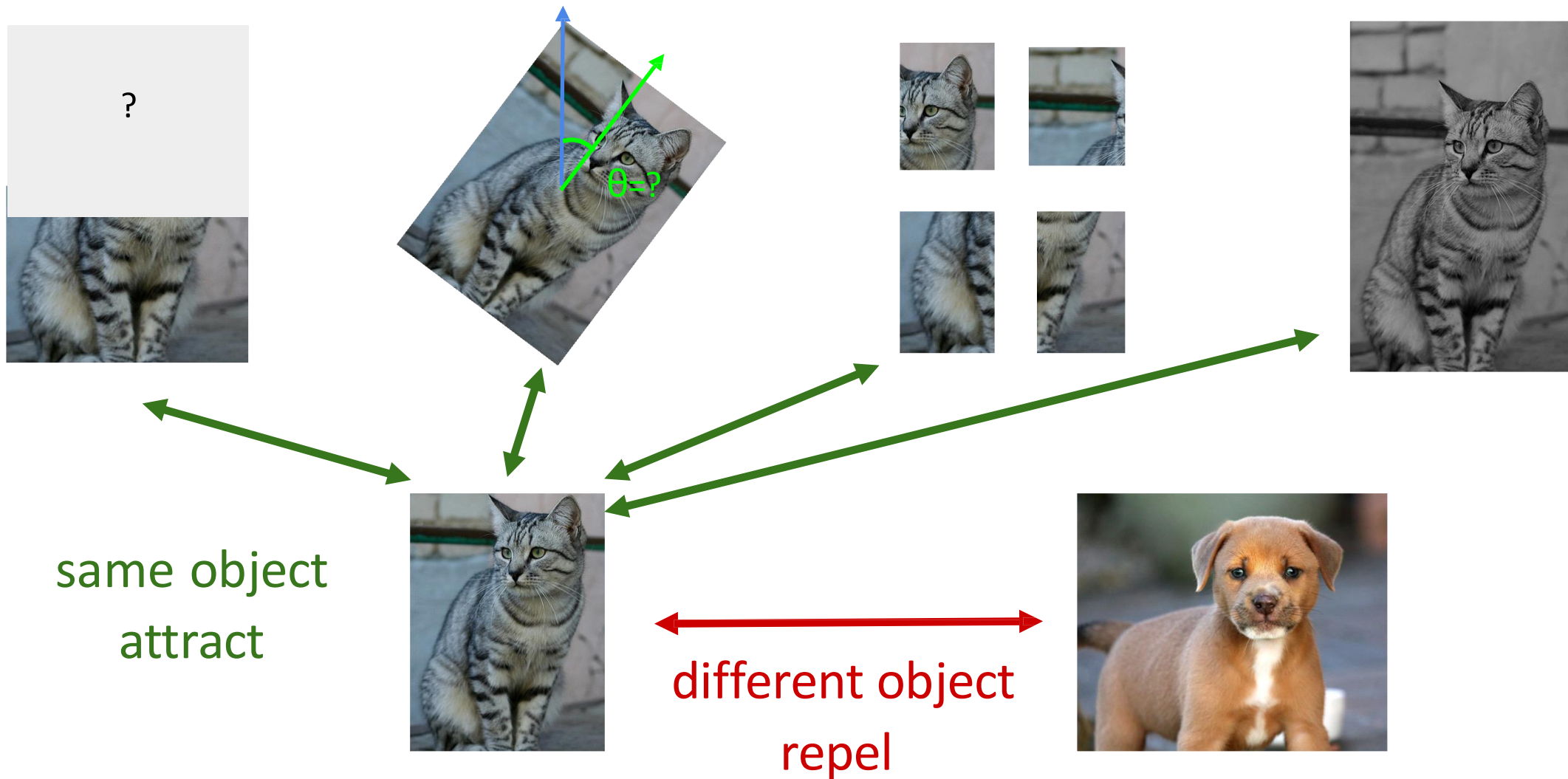
Trivial Solution

$$f_{\theta}(I) = \text{constant}$$

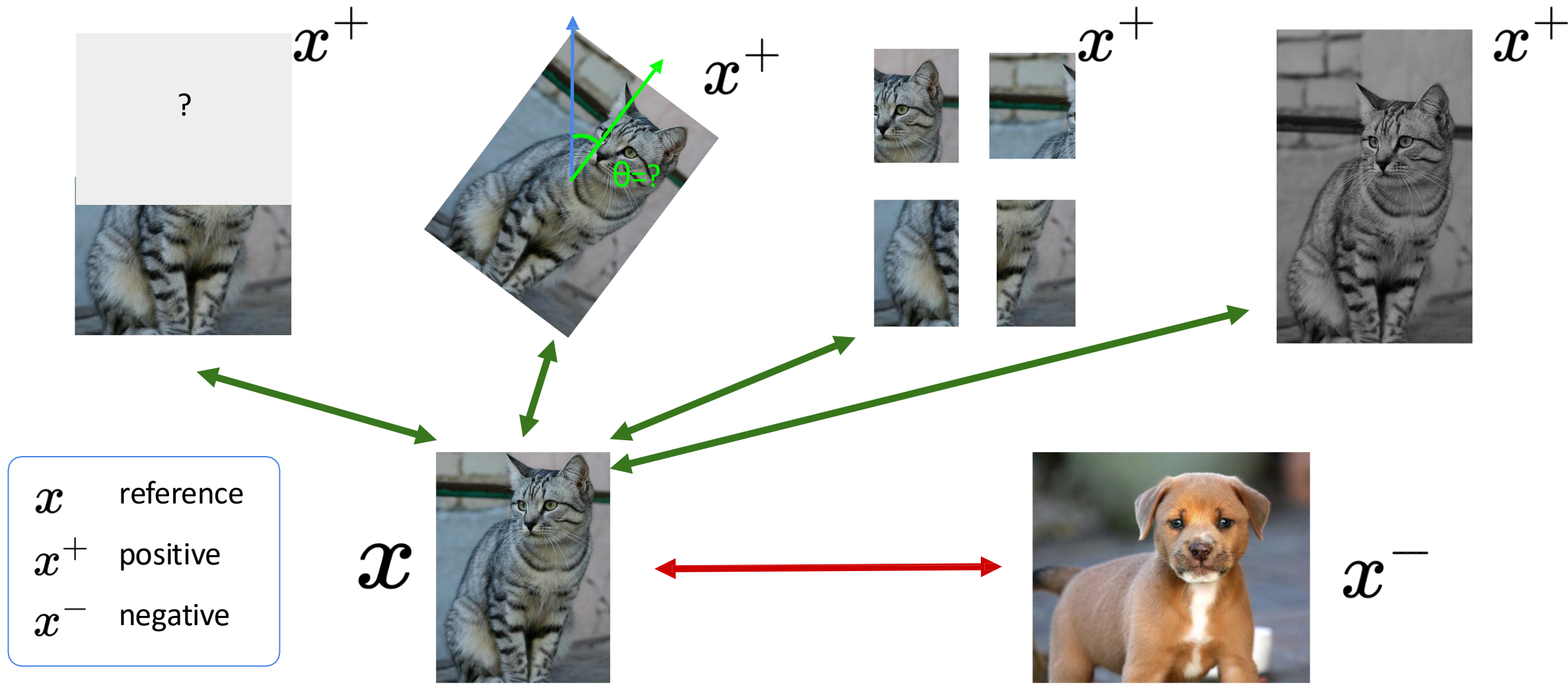
Satisfies the invariance property, but not useful
Often referred to as «representation collapse»

Contrastive learning methods allow to avoid this trivial solution

A more general pretext task?



Contrastive Representation Learning



A formulation of contrastive learning

What we want:

$$\text{score}(f(x), f(x^+)) \gg \text{score}(f(x), f(x^-))$$

x : reference sample; x^+ positive sample; x^- negative sample

Given a chosen score function, we aim to learn an encoder function f that yields high score for positive pairs (x, x^+) and low scores for negative pairs (x, x^-) .

A formulation of contrastive learning

Loss function given 1 positive sample and N - 1 negative samples:

$$L = -\mathbb{E}_X \left[\log \frac{\overbrace{\exp(s(f(x), f(x^+)))}}{\underbrace{\exp(s(f(x), f(x^+))) + \sum_{j=1}^{N-1} \exp(s(f(x), f(x_j^-)))} } \right]$$



x



x^+

Score for the positive pair



x



x_1^-



x_2^-



x_3^-

Scores for the $\ddots$ N-1 negative pairs

This seems familiar ...

Cross entropy loss for a N-way softmax classifier!

I.e., learn to find the positive sample from the N samples

SimCLR: A Simple Framework for Contrastive Learning

Cosine similarity as the score function:

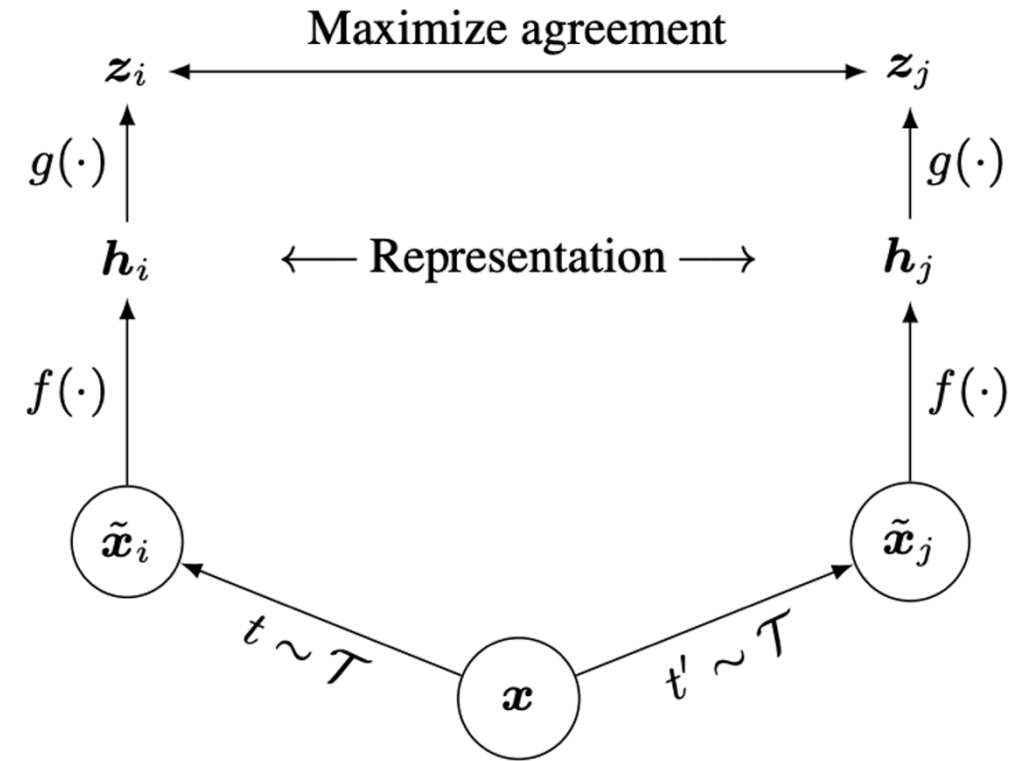
$$s(u, v) = \frac{u^T v}{||u|| ||v||}$$

Use a projection network $g(\cdot)$ to project features to a space where contrastive learning is applied

Generate positive samples through data augmentation:

- random cropping, random color distortion, and random blur.

Negatives are other images in the batch.



Source: [Chen et al., 2020](#)

SimCLR: generating positive samples from data augmentation



(a) Original



(b) Crop and resize



(c) Crop, resize (and flip)



(d) Color distort. (drop)



(e) Color distort. (jitter)



(f) Rotate $\{90^\circ, 180^\circ, 270^\circ\}$



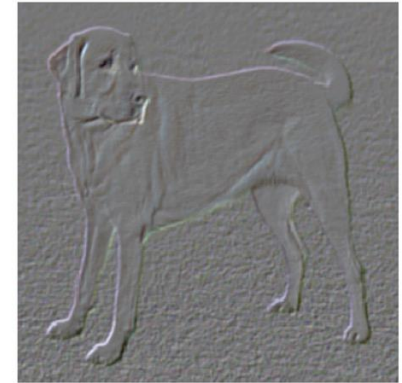
(g) Cutout



(h) Gaussian noise



(i) Gaussian blur

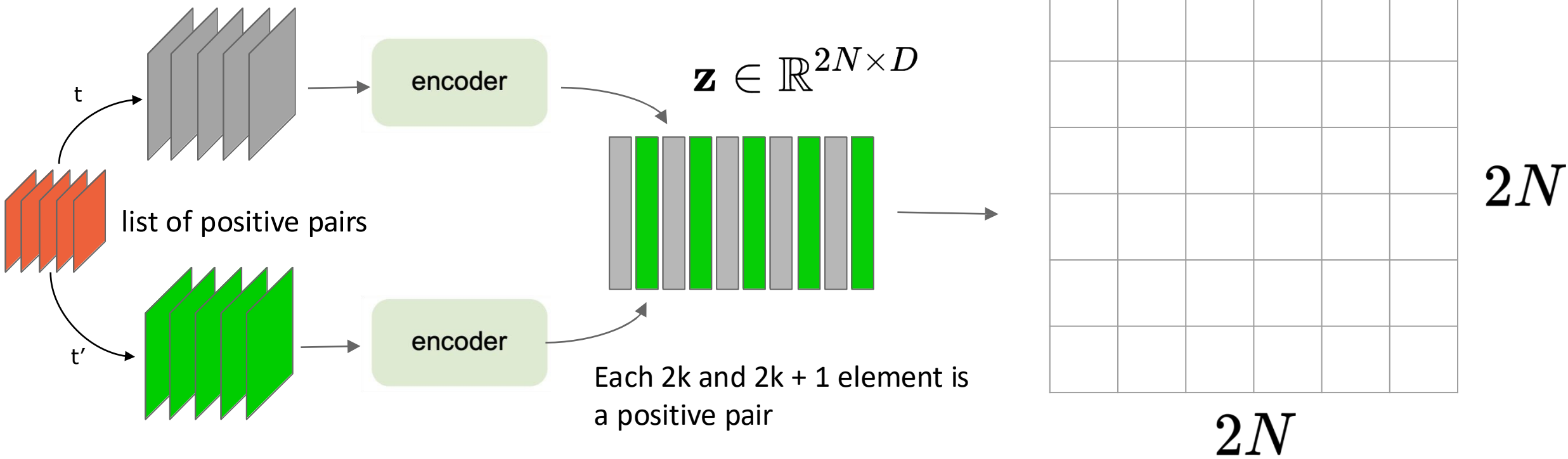


(j) Sobel filtering

Source: [Chen et al., 2020](#)

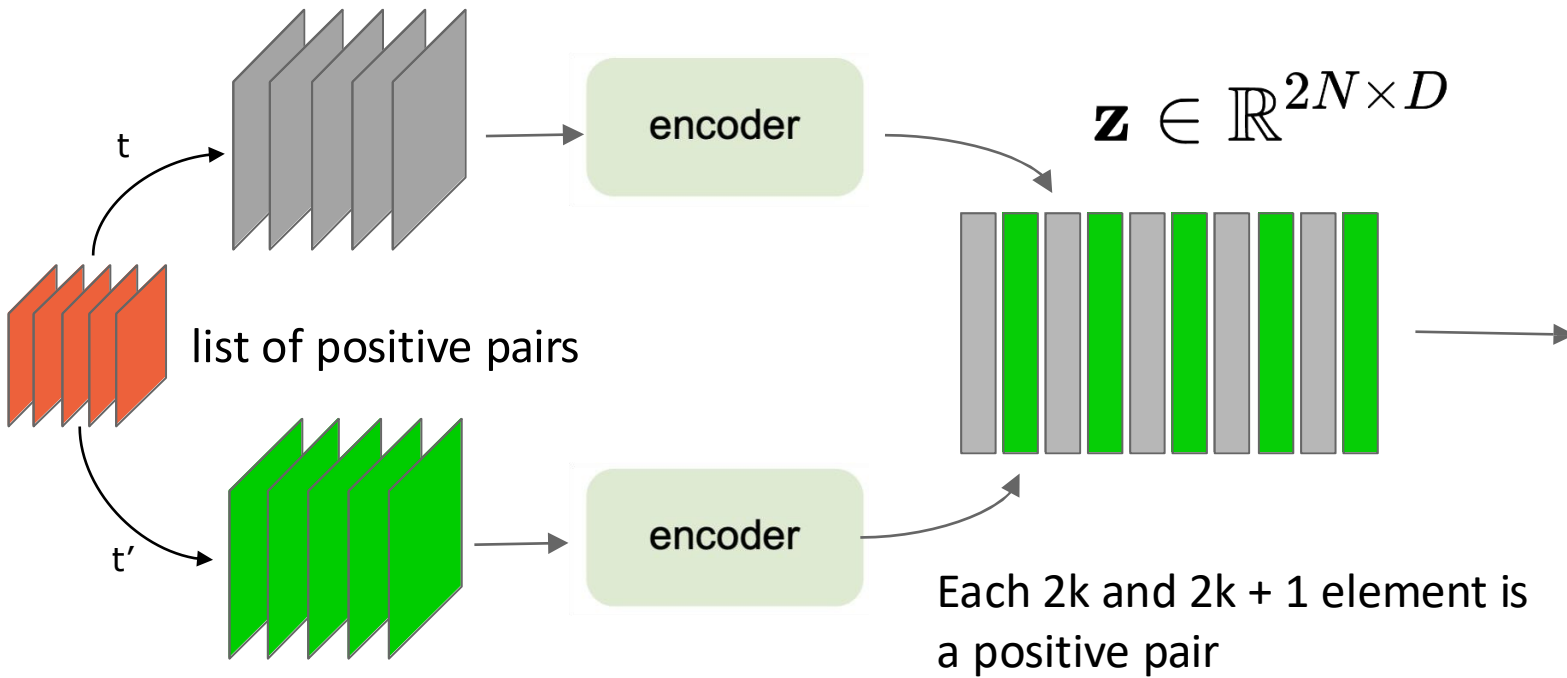
SimCLR: mini-batch training

transformations t and t' sampled randomly at each forward pass



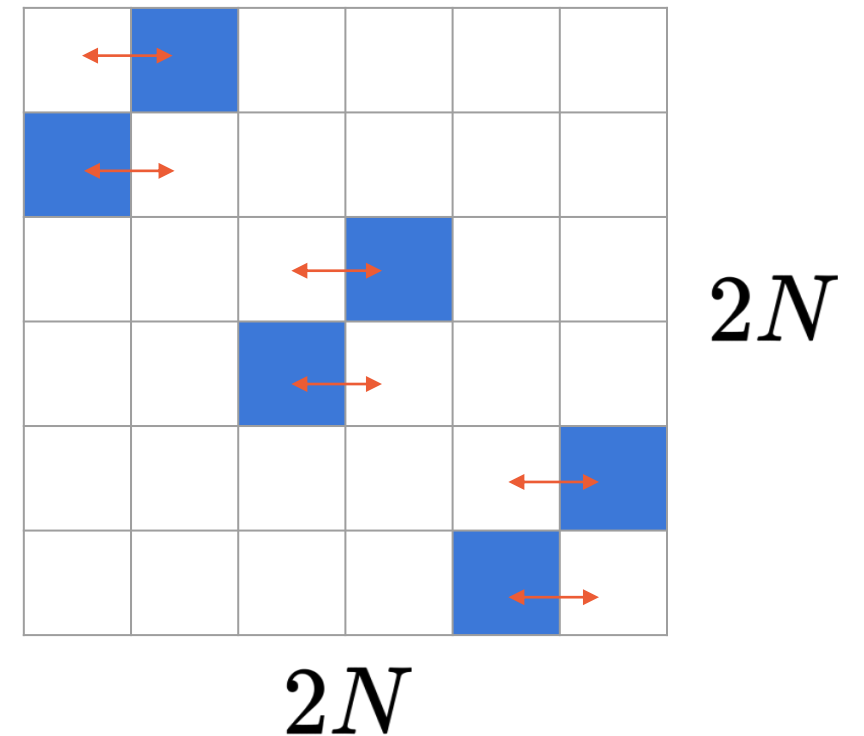
SimCLR: mini-batch training

transformations t and t' sampled randomly at each forward pass



$$s_{i,j} = \frac{z_i^T z_j}{||z_i|| ||z_j||}$$

"Affinity matrix"



 = classification label for each row

Semi-supervised learning on SimCLR features

Method	Architecture	Label fraction	
		1%	10%
		Top 5	
Supervised baseline	ResNet-50	48.4	80.4
<i>Methods using other label-propagation:</i>			
Pseudo-label	ResNet-50	51.6	82.4
VAT+Entropy Min.	ResNet-50	47.0	83.4
UDA (w. RandAug)	ResNet-50	-	88.5
FixMatch (w. RandAug)	ResNet-50	-	89.1
S4L (Rot+VAT+En. M.)	ResNet-50 (4×)	-	91.2
<i>Methods using representation learning only:</i>			
InstDisc	ResNet-50	39.2	77.4
BigBiGAN	RevNet-50 (4×)	55.2	78.8
PIRL	ResNet-50	57.2	83.8
CPC v2	ResNet-161(*)	77.9	91.2
SimCLR (ours)	ResNet-50	75.5	87.8
SimCLR (ours)	ResNet-50 (2×)	83.0	91.2
SimCLR (ours)	ResNet-50 (4×)	85.8	92.6

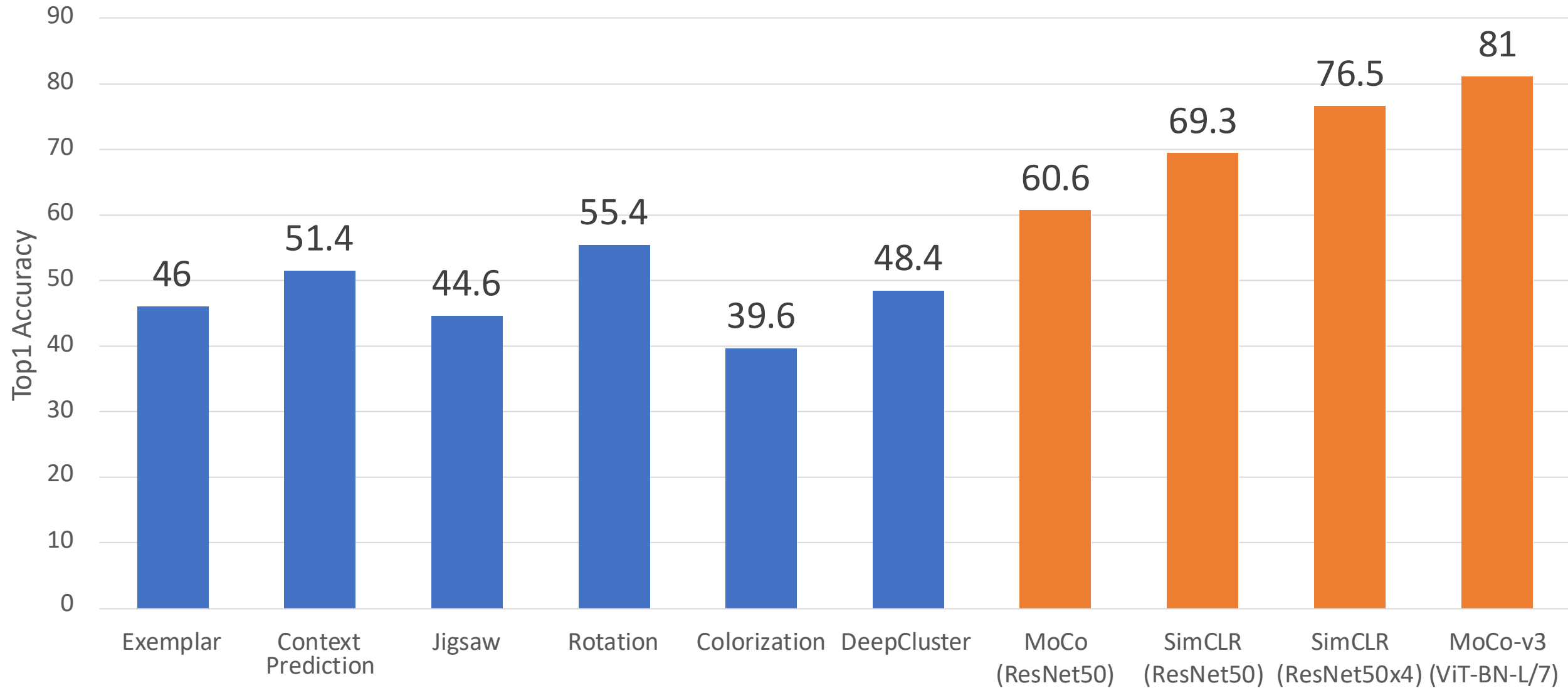
Table 7. ImageNet accuracy of models trained with few labels.

Train feature encoder on ImageNet (entire training set) using SimCLR.

Finetune the encoder with 1% / 10% of labeled data on ImageNet.

Source: [Chen et al., 2020](#)

ImageNet Linear Classification from SSL Features



He et al, "Momentum Contrast for Unsupervised Visual Representation Learning", CVPR 2020
Chen et al, "A Simple Framework for Contrastive Learning of Visual Representations", ICML 2020
Chen et al, "An Empirical Study of Training Self-Supervised Vision Transformers", ICCV 2021

(Lots of caveats here ... different architectures, etc)

Negative-free Contrastive Methods

The problem with negatives...

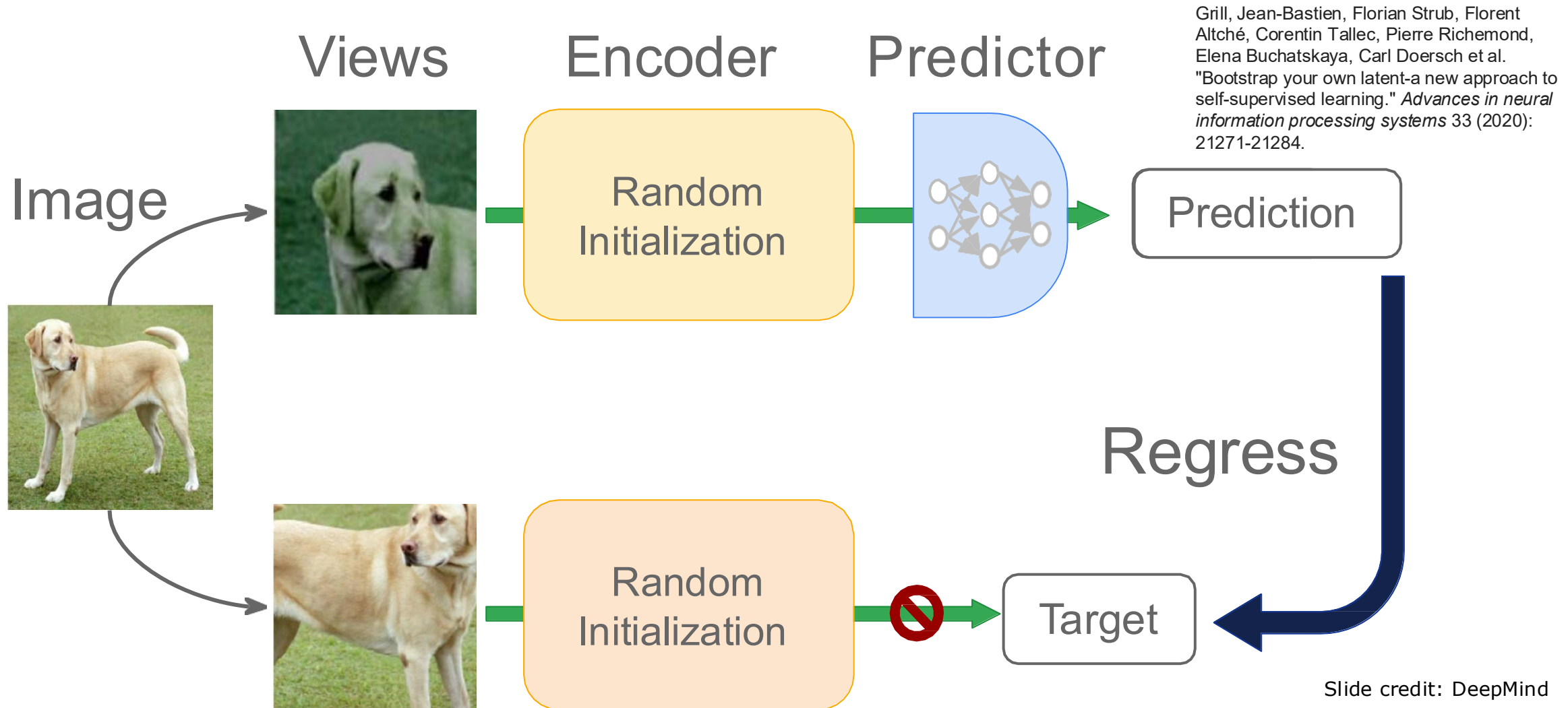
The use of negatives in contrastive learning is powerful, but it comes with problems and trade-offs:

- Sampling Bias:
 - Not all negatives are truly dissimilar
 - In large datasets, negatives may include samples from the same class or semantically similar content
 - This causes false negatives, which hurt learning – the model is penalized for pulling together things that should be similar
 - *Example:* Two clips of people swimming — sampled from different videos but semantically the same action — are treated as negatives.
- Requires Large Batches or Memory Banks
 - InfoNCE converges to the Mutual Information lowerbound for large N
 - This leads to the need for large batches
 - If there are not enough negatives, representations may collapse
- Negatives make methods difficult to tune
 - Methods are sensitive to the augmentation strategy used to produce positives

Bootstrap Your Own Latent (BYOL)

Idea: train two separate encoders independently with no negatives

Since the encoders are not trained jointly, they will not collapse to same representation for all frames

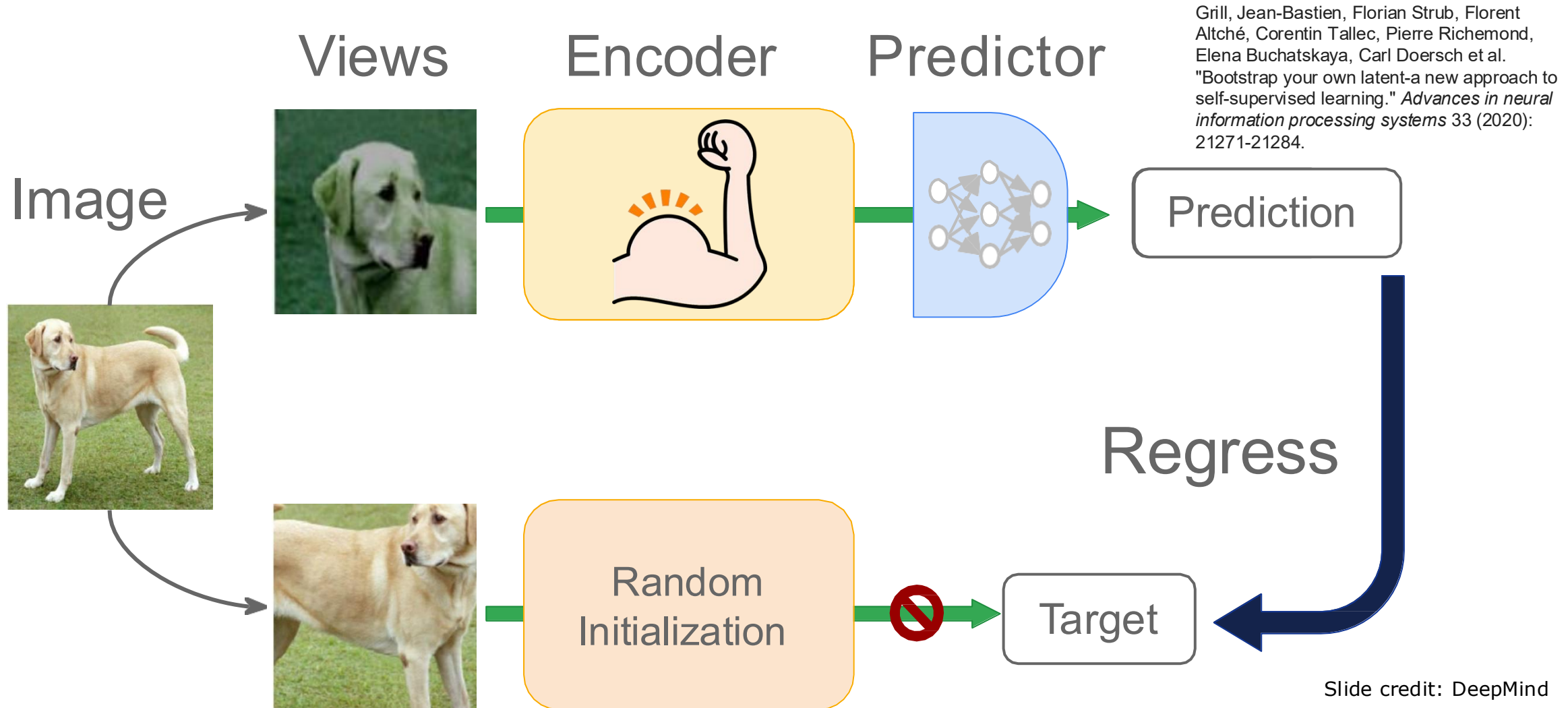


Slide credit: DeepMind

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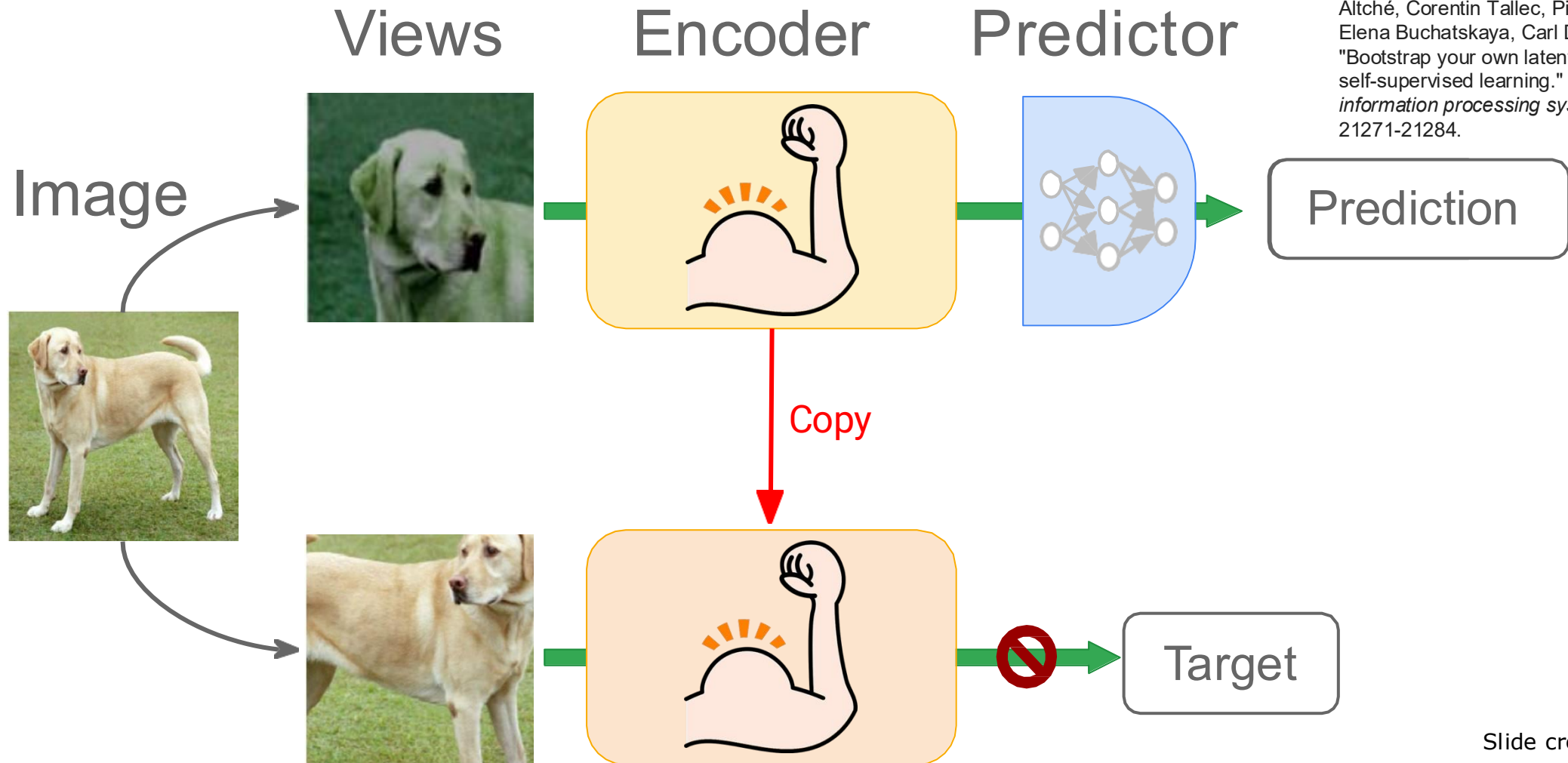


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Grill, Jean-Bastien, Florian Strub, Florent Altché, Corentin Tallec, Pierre Richemond, Elena Buchatskaya, Carl Doersch et al.
"Bootstrap your own latent-a new approach to self-supervised learning." *Advances in neural information processing systems* 33 (2020): 21271-21284.

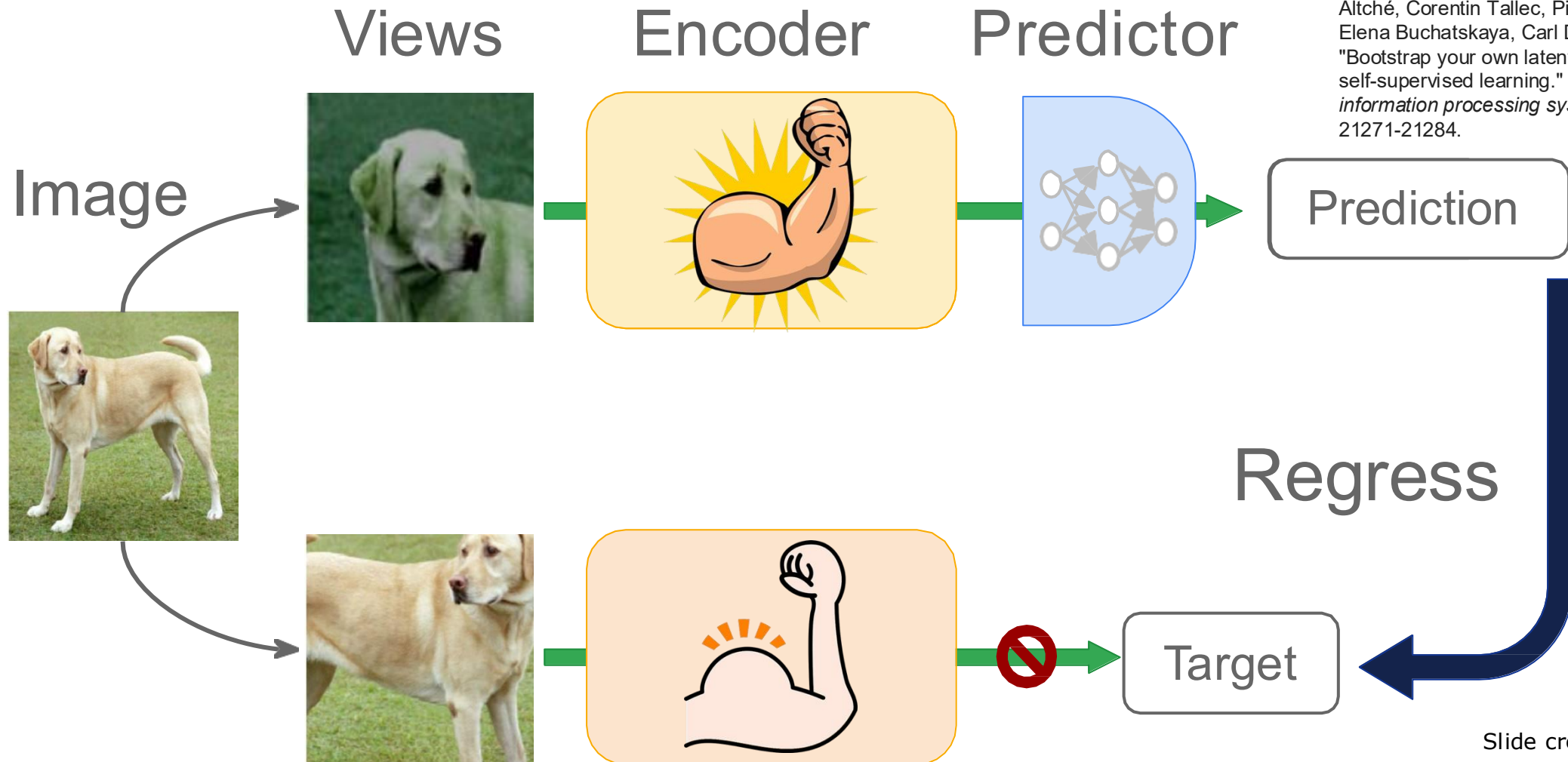


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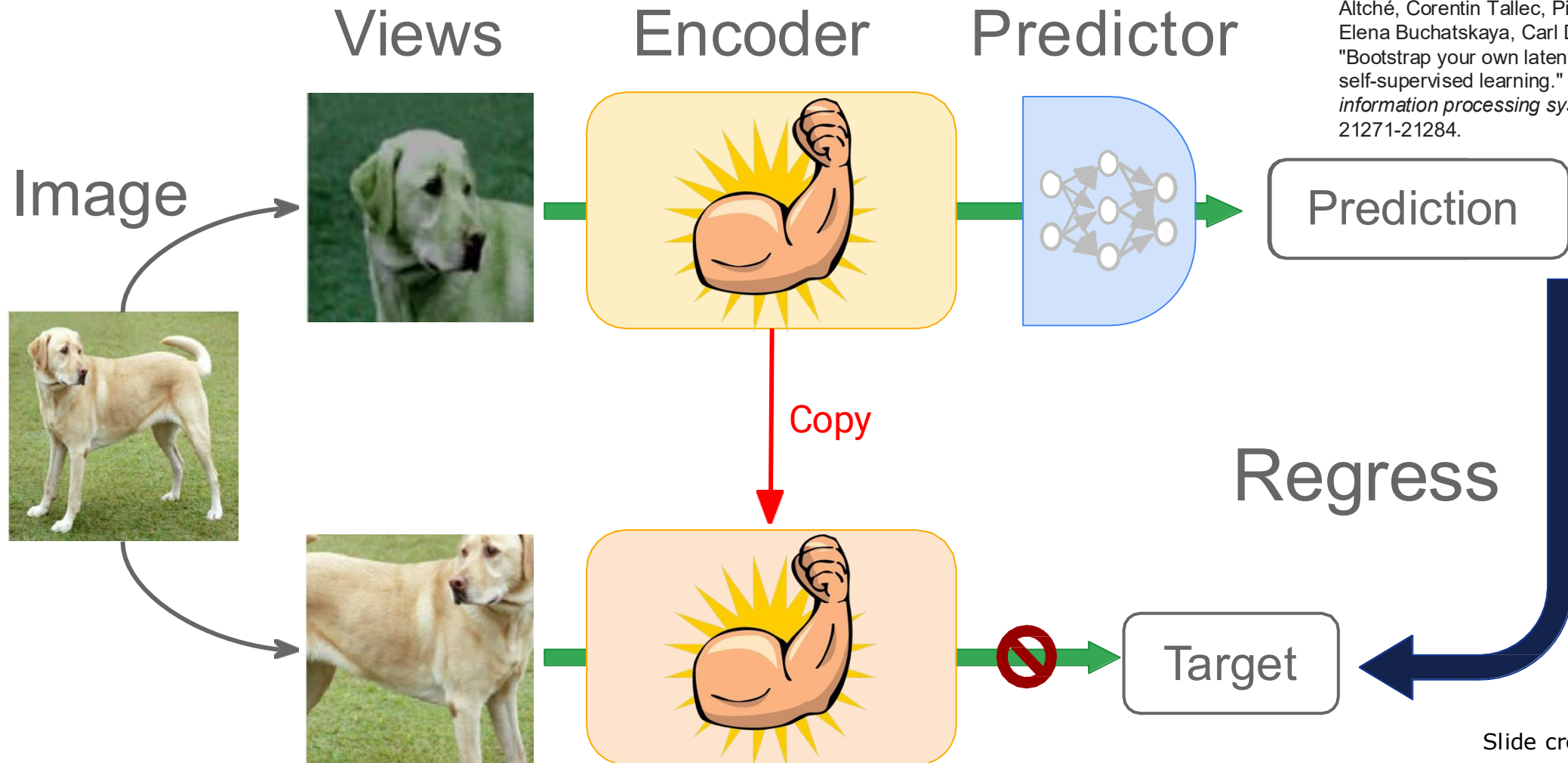
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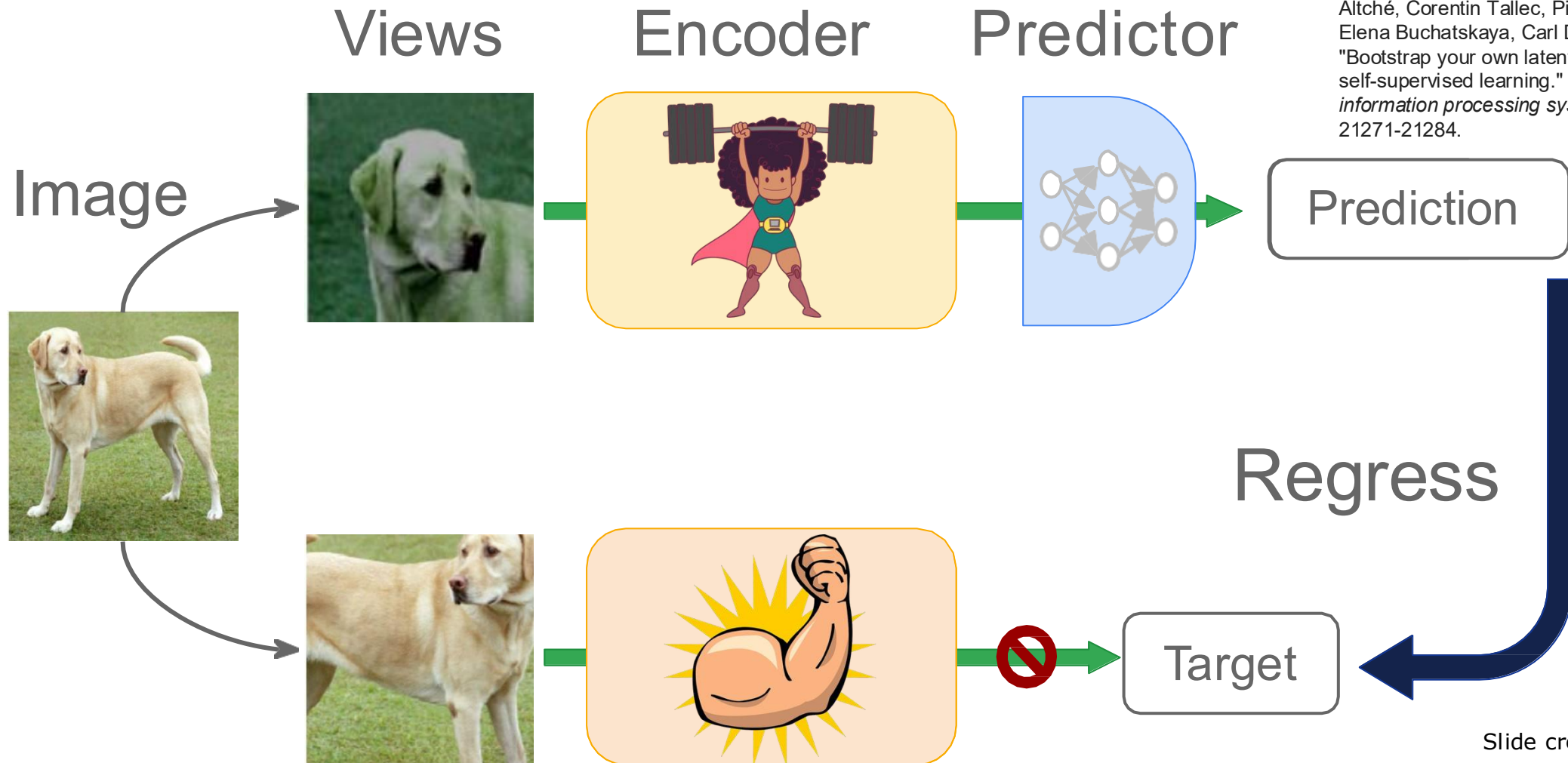
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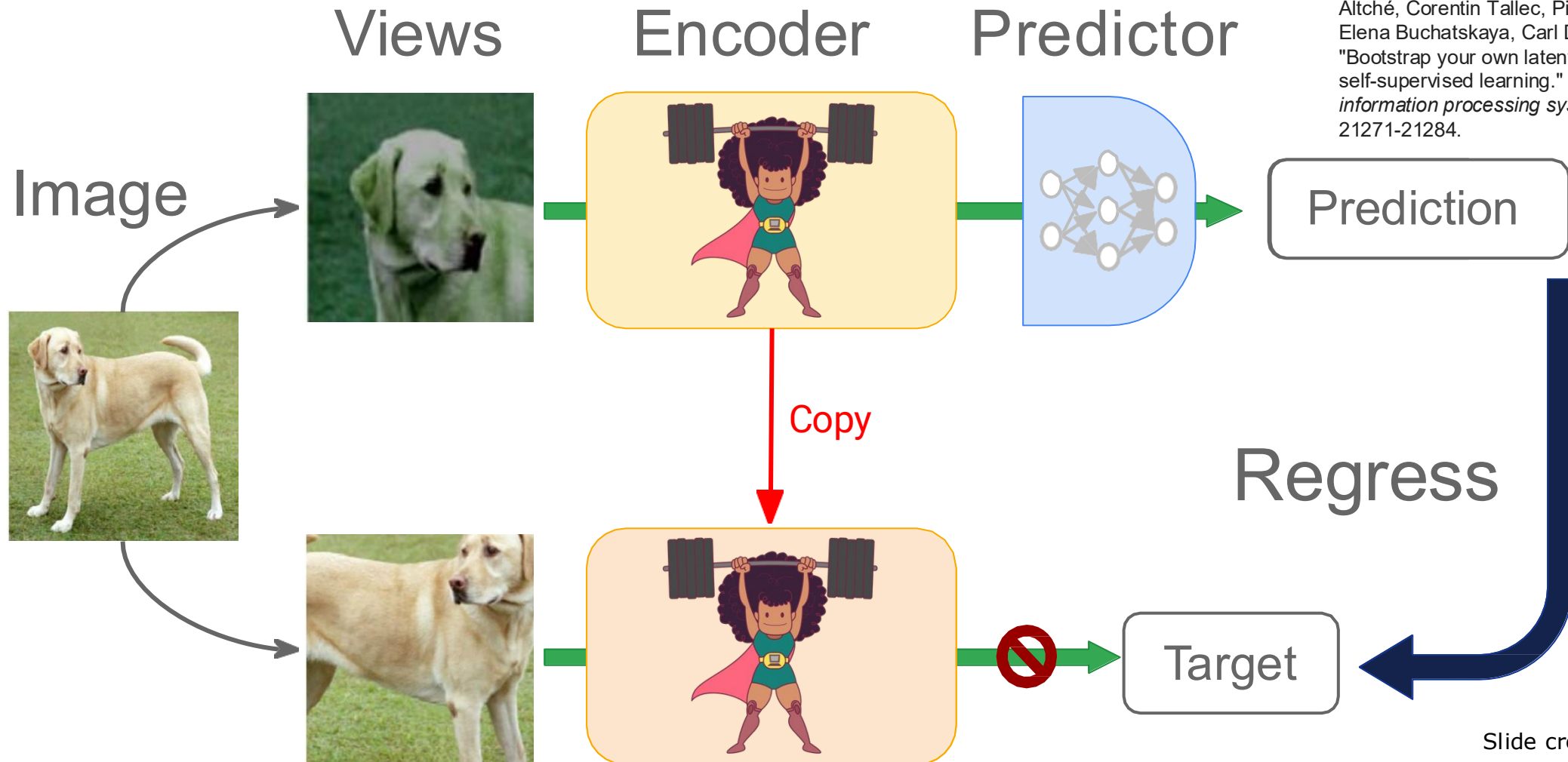
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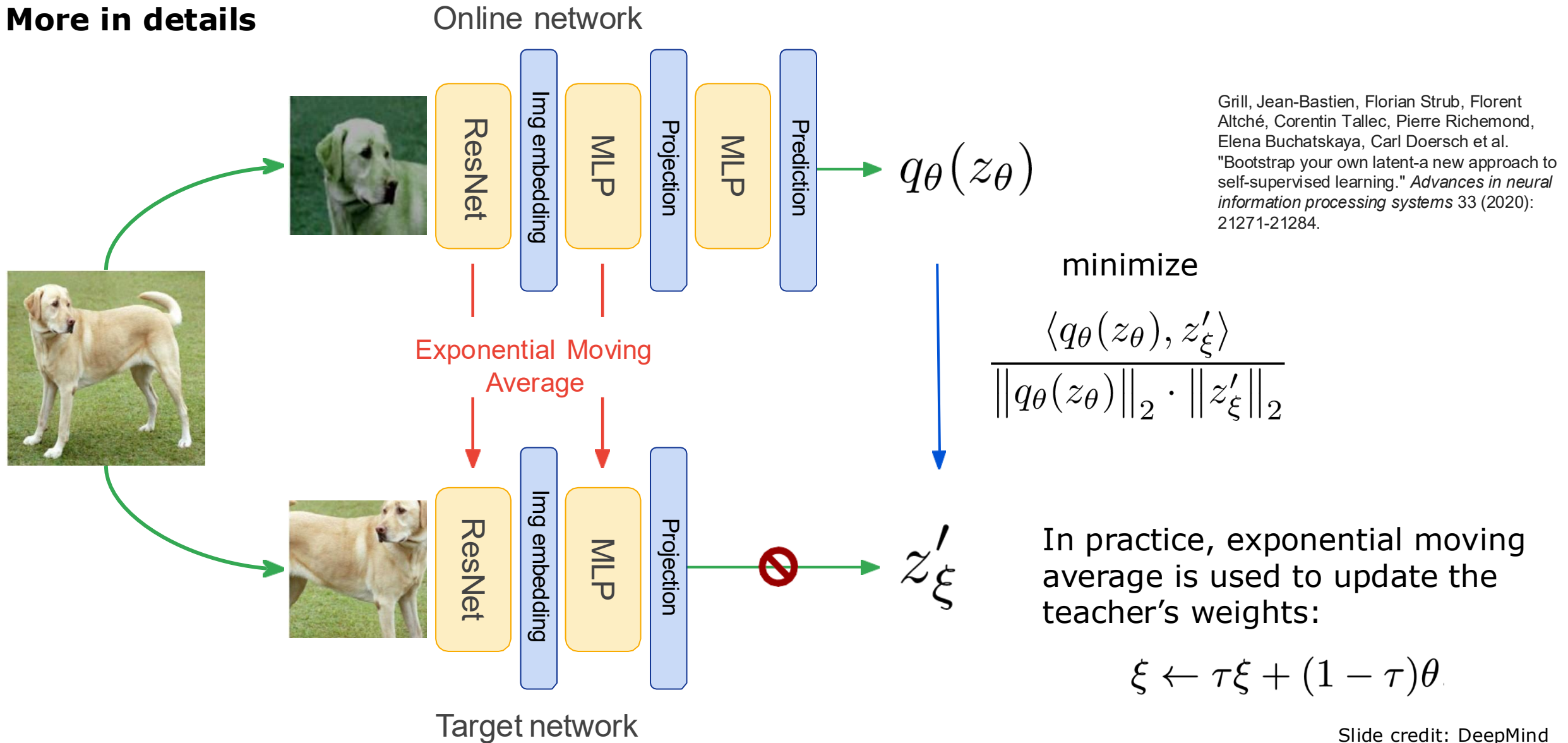
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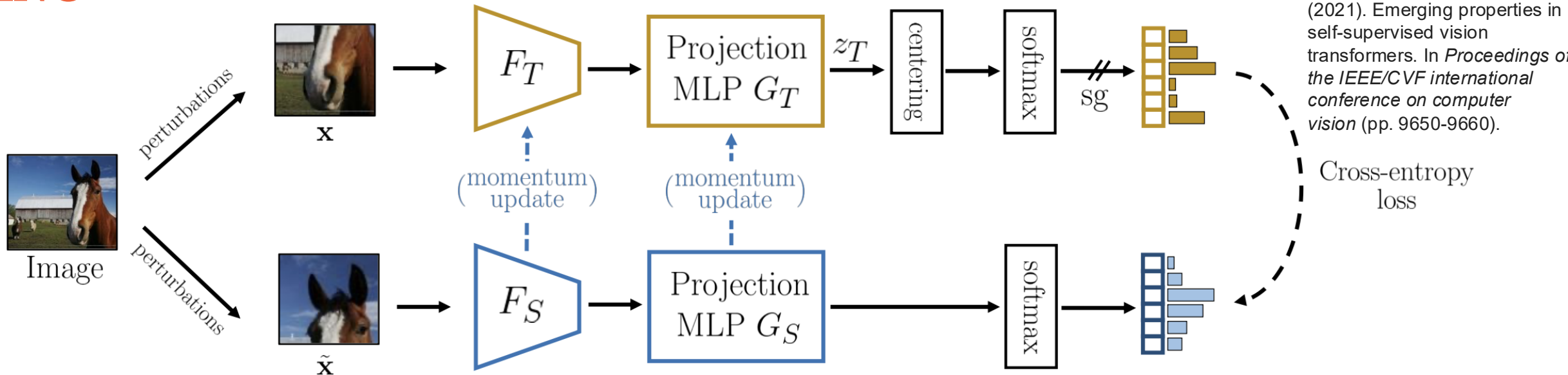
Slide credit: DeepMind

BYOL Architecture

More in details



Grill, Jean-Bastien, Florian Strub, Florent Altché, Corentin Tallec, Pierre Richemond, Elena Buchatskaya, Carl Doersch et al. "Bootstrap your own latent-a new approach to self-supervised learning." *Advances in neural information processing systems* 33 (2020): 21271-21284.



Caron, M., Touvron, H., Misra, I., Jégou, H., Mairal, J., Bojanowski, P., & Joulin, A. (2021). Emerging properties in self-supervised vision transformers. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 9650-9660).

DINO extends over BYOL and previous works with some key changes:

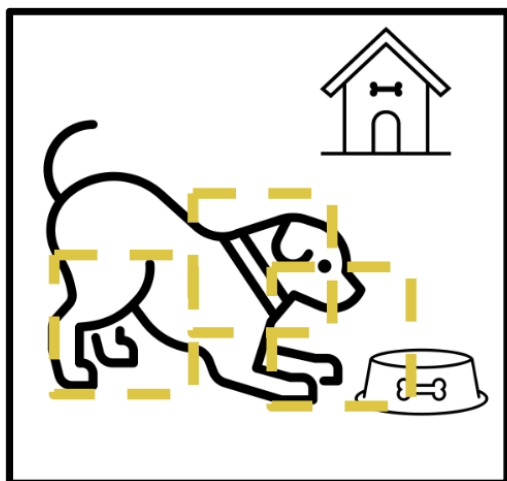
- No prediction head post-processing
- Features are centered (subtract running mean of teacher's output) and passed through a softmax with low temperature – this creates a sharp distribution which is used as a pseudo-label
- A cross-entropy loss is used to train the student, rather than MSE

Method	Mom.	Loss	Pred.	k -NN	Lin.
DINO	✓	CE	✗	72.8	76.1
	✗	CE	✗	0.1	0.1
	✓	MSE	✗	52.6	62.4
	✓	CE	✓	71.8	75.6
BYOL	✓	MSE	✓	66.6	71.4

DINO Details – Multi-Crop Strategy

Caron, M., Touvron, H., Misra, I., Jégou, H., Mairal, J., Bojanowski, P., & Joulin, A. (2021). Emerging properties in self-supervised vision transformers. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 9650-9660).

Local crops



Global crops



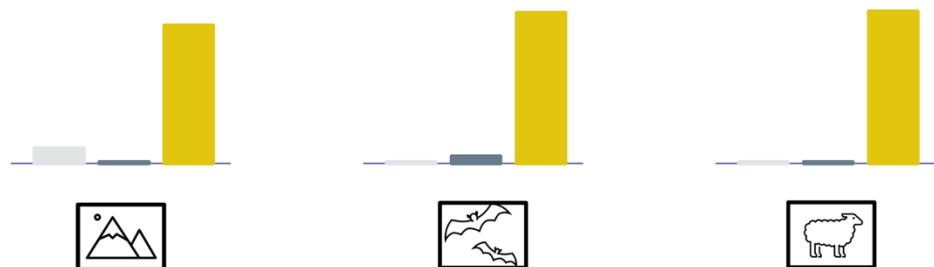
Local predict the pseudo-label of global

Local-to-global matching

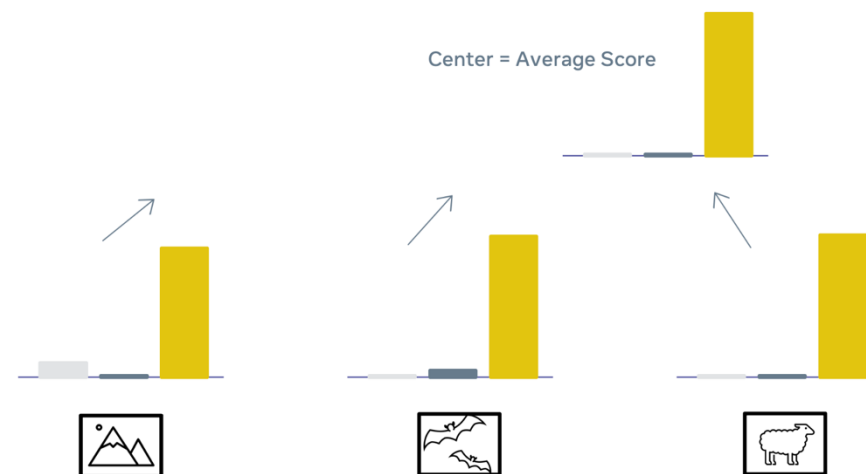
A multi-crop strategy is applied, where a «local-to-global» matching is learned.

DINO Details – Centering and Sharpening

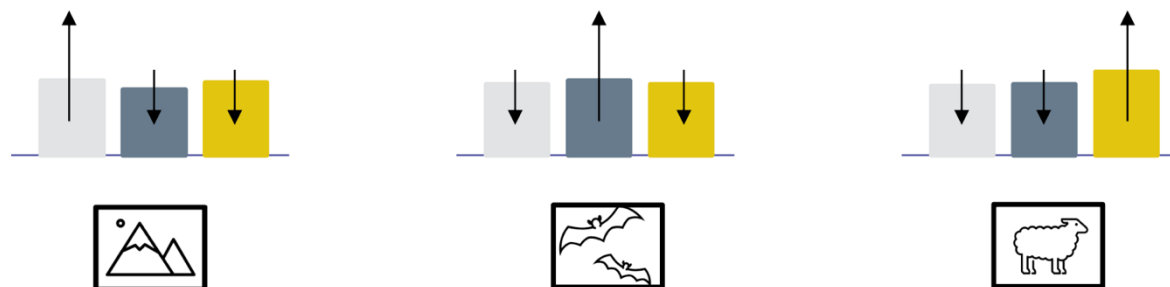
Some dimensions in the embedding space might have a wider scale, hence dominating the distribution of activations.



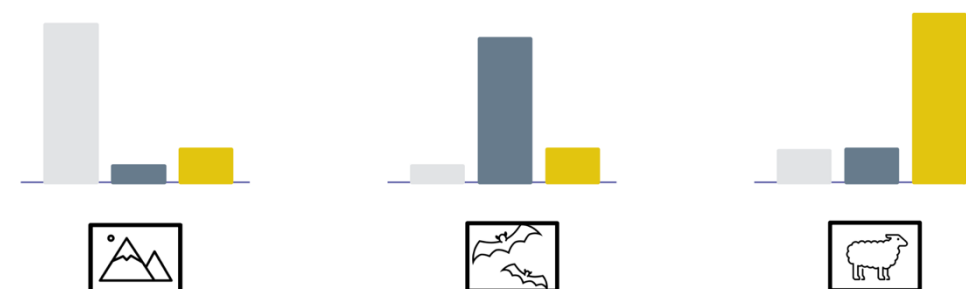
1) Original Features



2) Computing Average (running)



3) Centered Features (average subtracted)



4) Sharpening (low temperature softmax)

DINOv2: Learning Robust Visual Features without Supervision

Maxime Oquab**, Timothée Darcet**, Théo Moutakanni**,
Huy V. Vo*, Marc Szafraniec*, Vasil Khalidov*, Pierre Fernandez, Daniel Haziza,
Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba,
Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat,
Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jegou, Julien Mairal¹,
Patrick Labatut*, Armand Joulin*, Piotr Bojanowski*

Meta AI Research ¹*Inria*

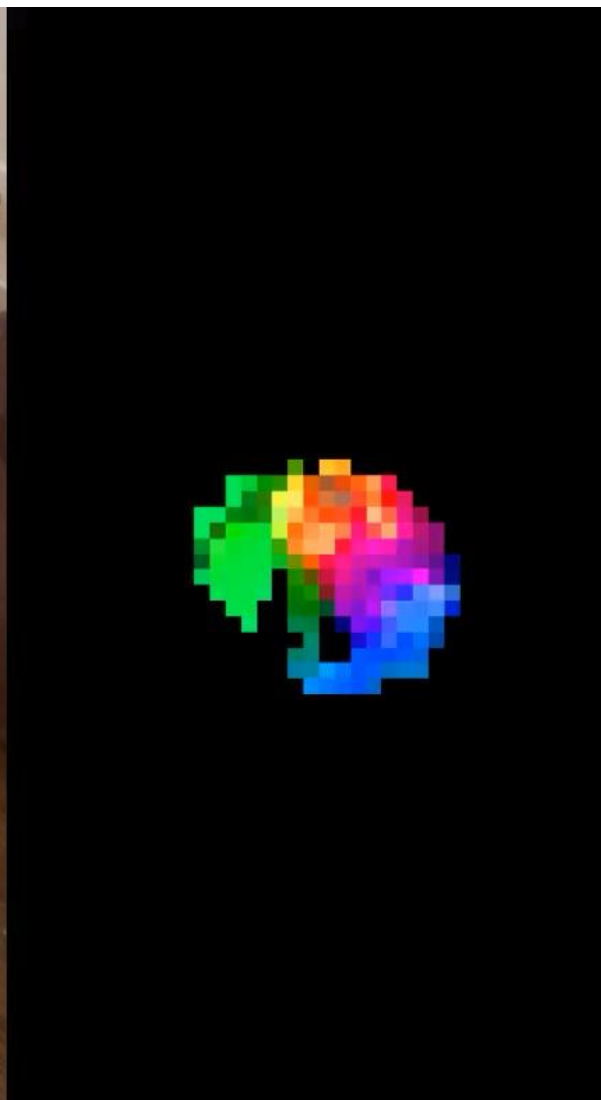
*core team **equal contribution

Reviewed on OpenReview: <https://openreview.net/forum?id=a68SUt6zFt>

Oquab, Maxime, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez et al. "Dinov2: Learning robust visual features without supervision." *arXiv preprint arXiv:2304.07193* (2023).

DINOv2 – Emerging Properties

<https://dinov2.metademolab.com>



DINOv2 – Emerging Properties

<https://dinov2.metademolab.com>



Dense Matching

DINOv3

Oriane Siméoni* Huy V. Vo* Maximilian Seitzer* Federico Baldassarre* Maxime Oquab*
Cijo Jose Vasil Khalidov Marc Szafraniec Seungeun Yi Michaël Ramamonjisoa
Francisco Massa Daniel Haziza Luca Wehrstedt Jianyuan Wang
Timothée Darcet Théo Moutakanni Leonel Sentana Claire Roberts
Andrea Vedaldi Jamie Tolan John Brandt¹ Camille Couprie
Julien Mairal² Hervé Jégou Patrick Labatut Piotr Bojanowski

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<https://ai.meta.com/research/dinov3/>

Reconstructive Approaches

Masked Auto-Encoding

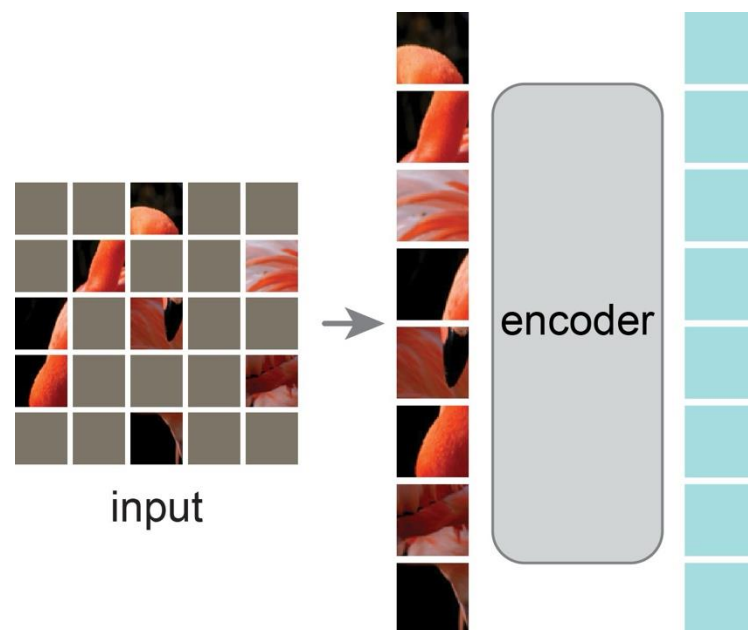
He, K., Chen, X., Xie, S., Li, Y., Dollár, P., & Girshick, R. (2022). Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 16000-16009).



Random masking

Masked Auto-Encoding

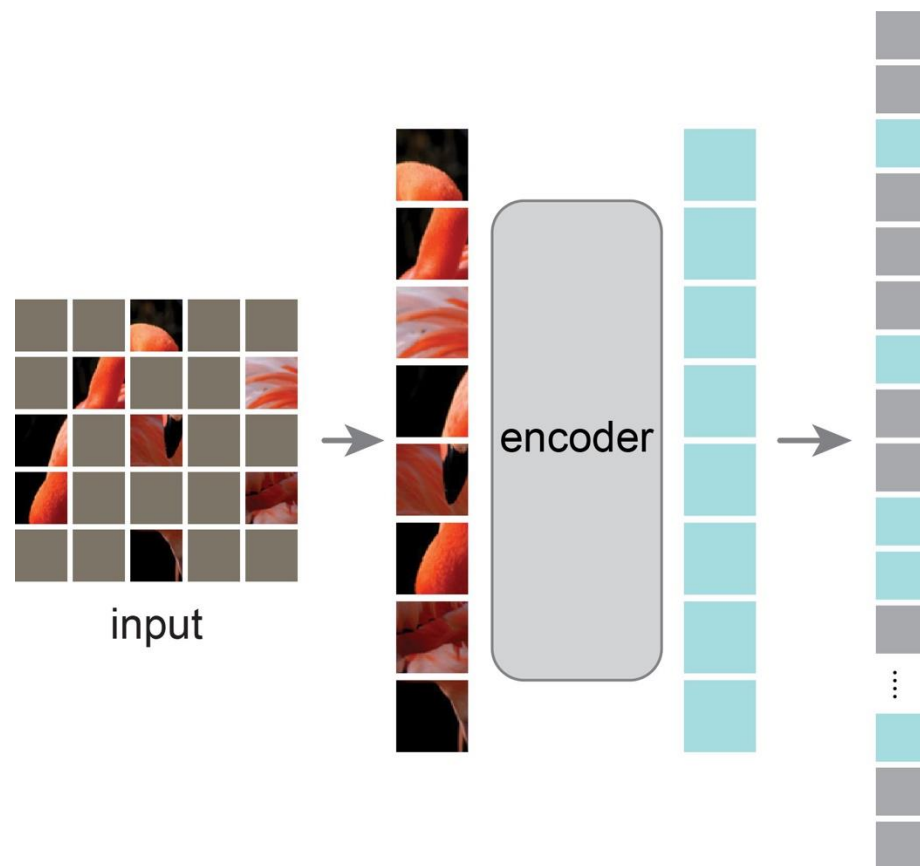
He, K., Chen, X., Xie, S., Li, Y., Dollár, P., & Girshick, R. (2022). Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 16000-16009).



Encode visible patches

Masked Auto-Encoding

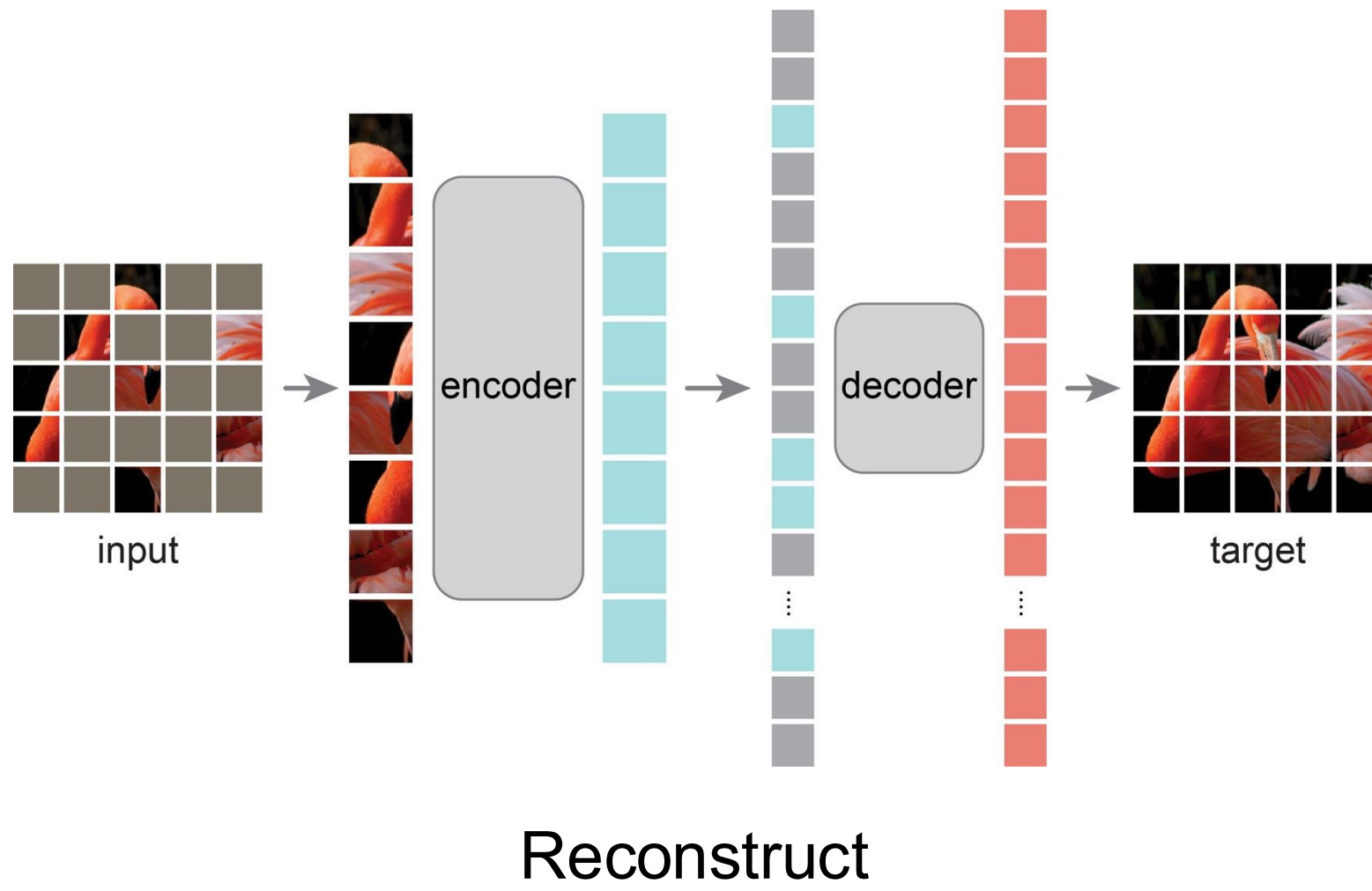
He, K., Chen, X., Xie, S., Li, Y., Dollár, P., & Girshick, R. (2022). Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 16000-16009).



Add mask tokens

Masked Auto-Encoding

He, K., Chen, X., Xie, S., Li, Y., Dollár, P., & Girshick, R. (2022). Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 16000-16009).

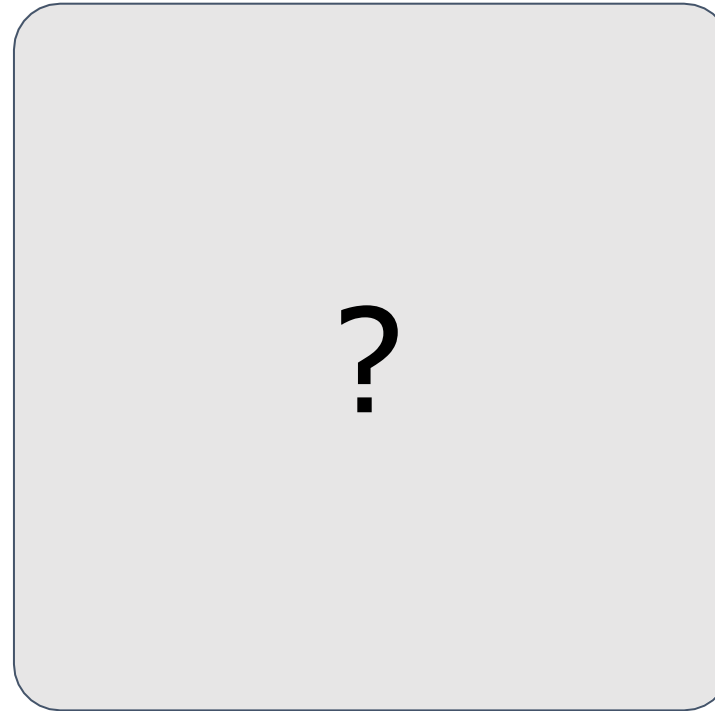


Reconstruction Error

He, K., Chen, X., Xie, S., Li, Y., Dollár, P., & Girshick, R. (2022). Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 16000-16009).



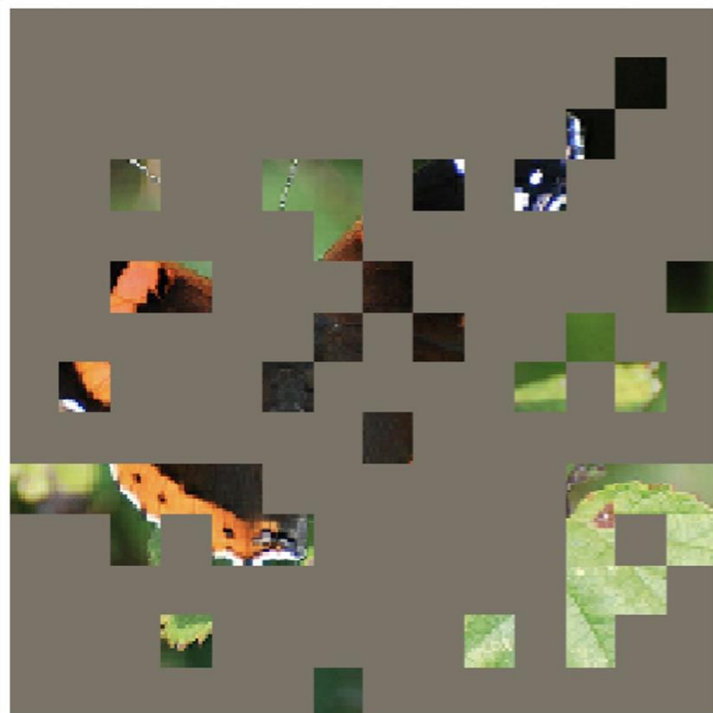
Masked input: 80%



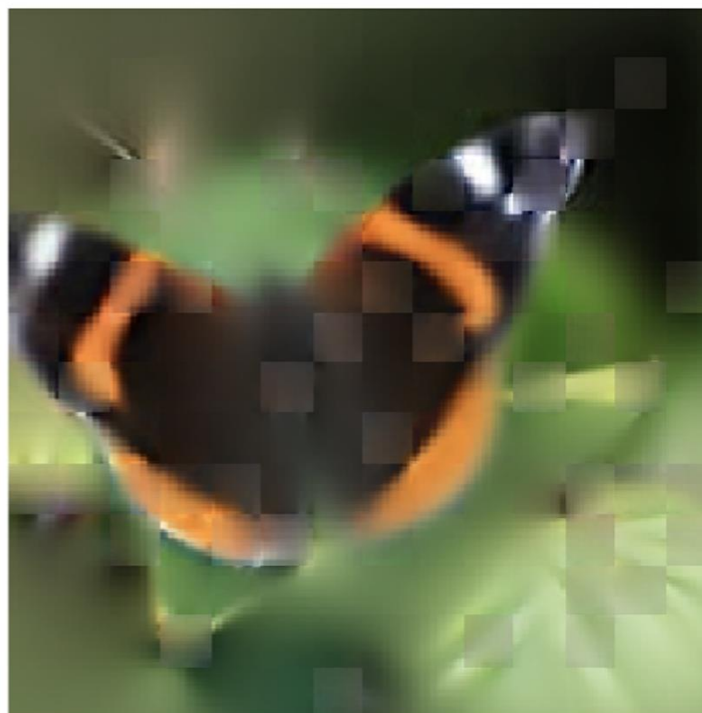
You guess?

Reconstruction Error

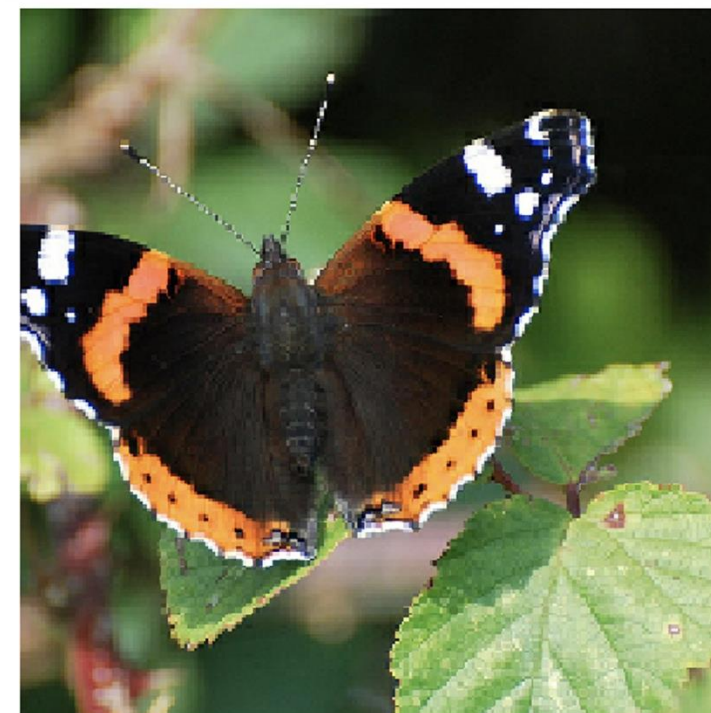
He, K., Chen, X., Xie, S., Li, Y., Dollár, P., & Girshick, R. (2022). Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 16000-16009).



Masked input: 80%



MAE's guess

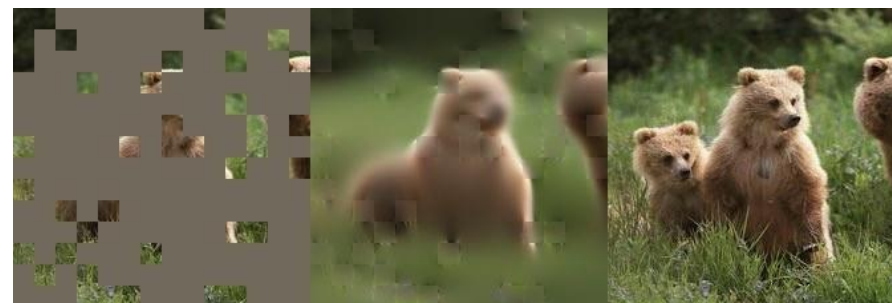
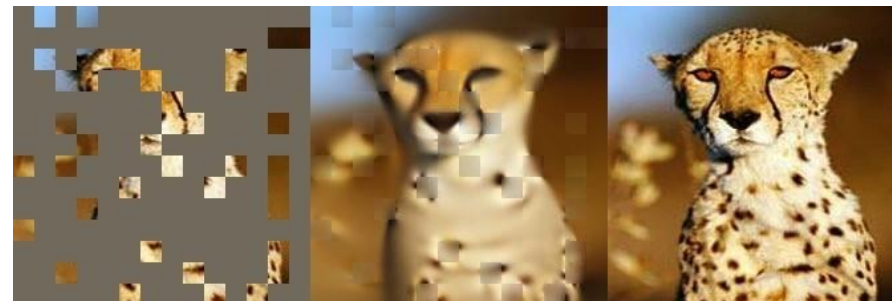
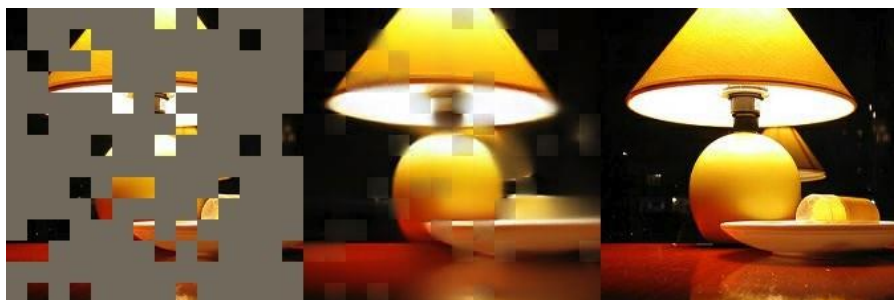
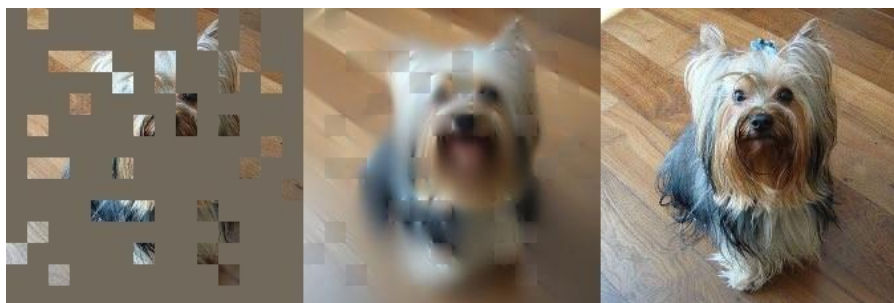


Ground truth

Reconstruction Error

ImageNet val set (unseen)

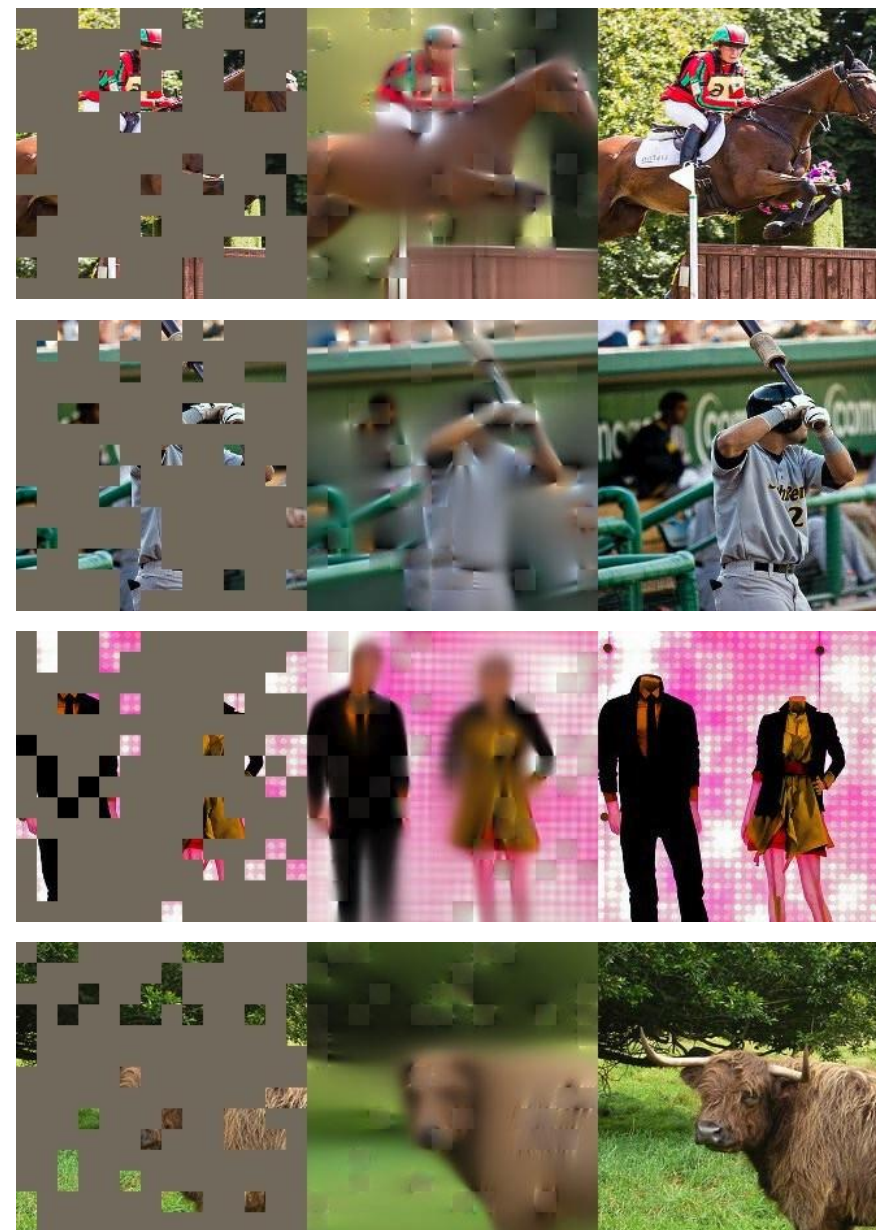
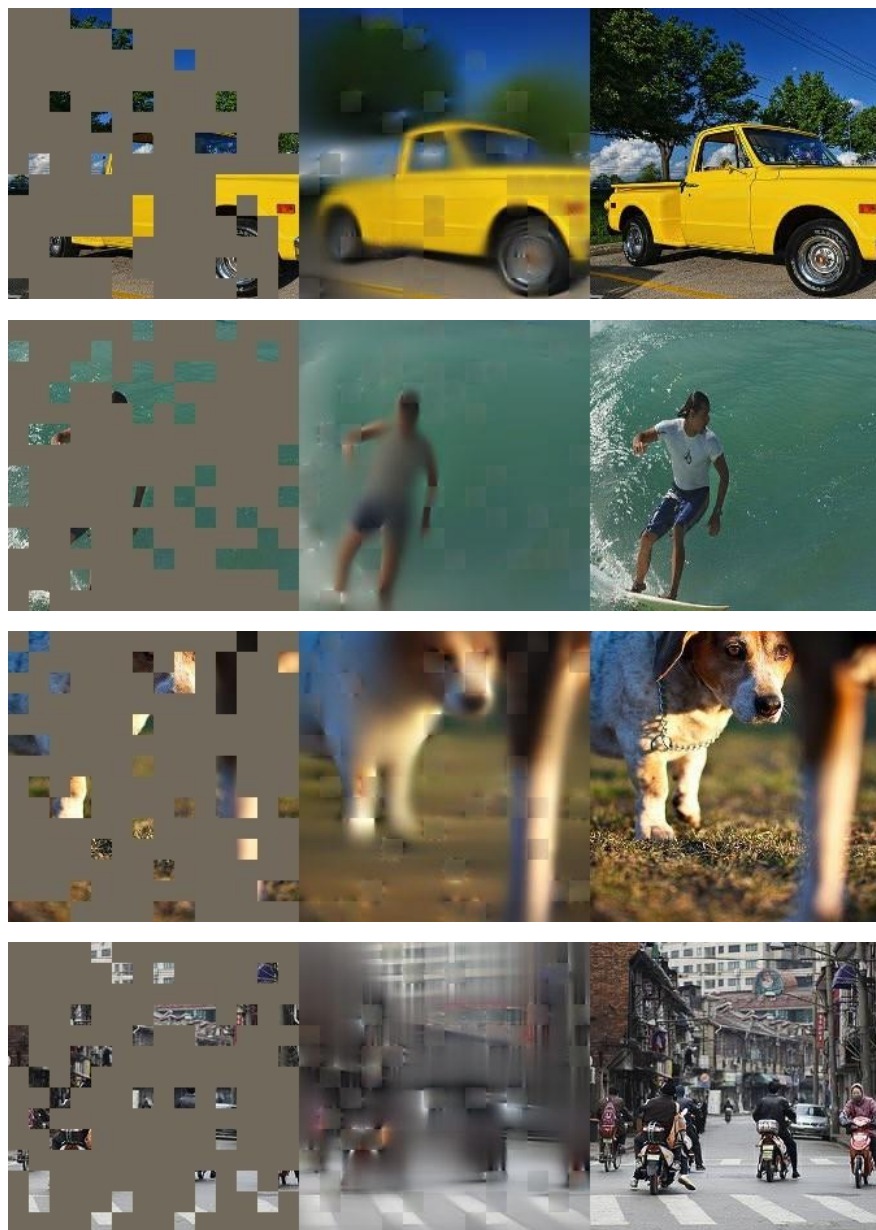
Uni
ct



Reconstruction Error

ImageNet val set (unseen)

Uni
ct



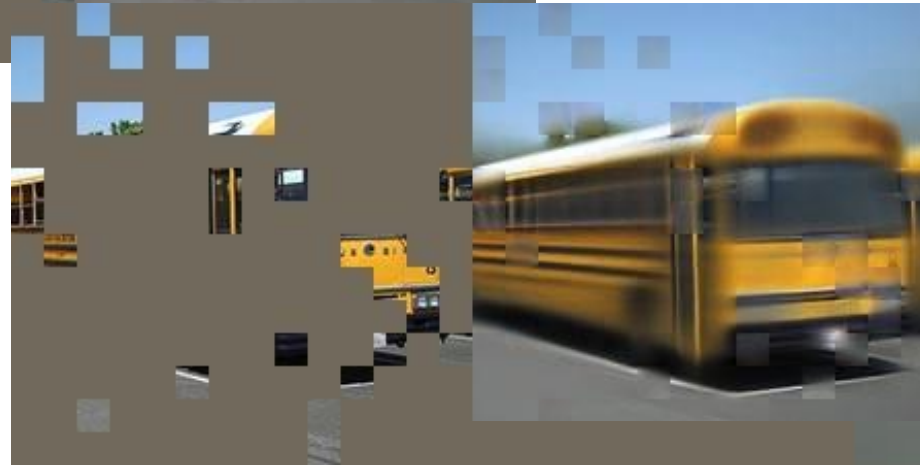
MAE Can Generalize

75% mask

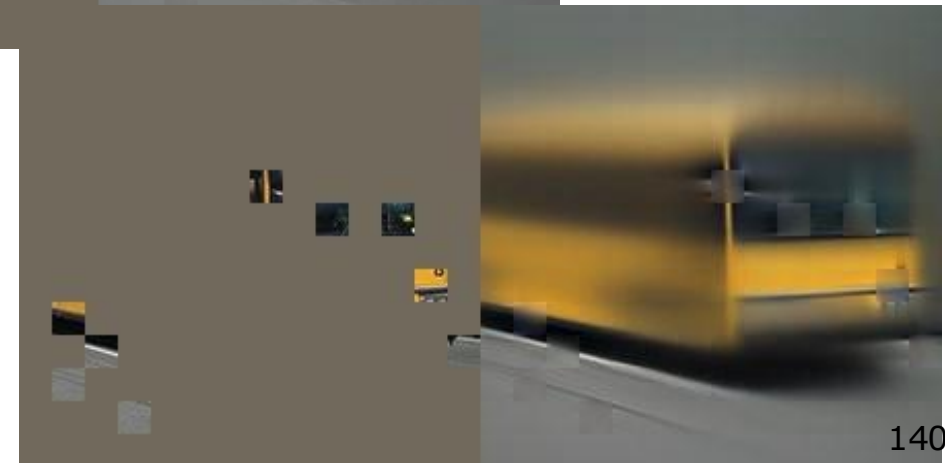


original

85% mask

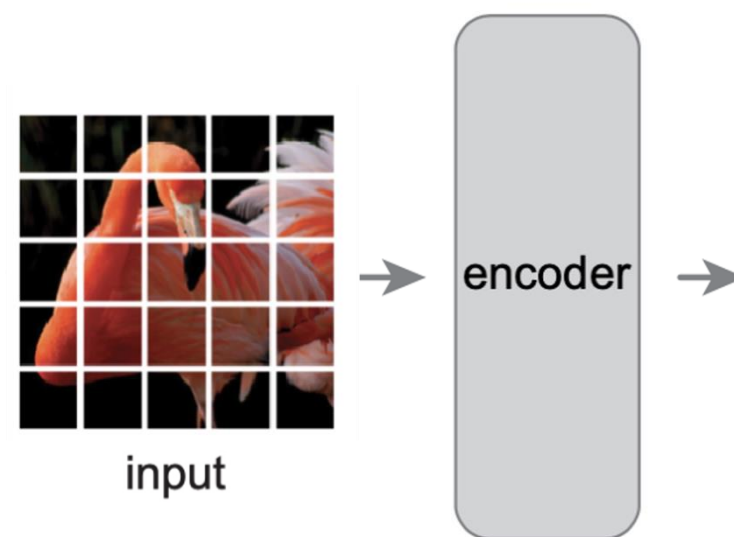


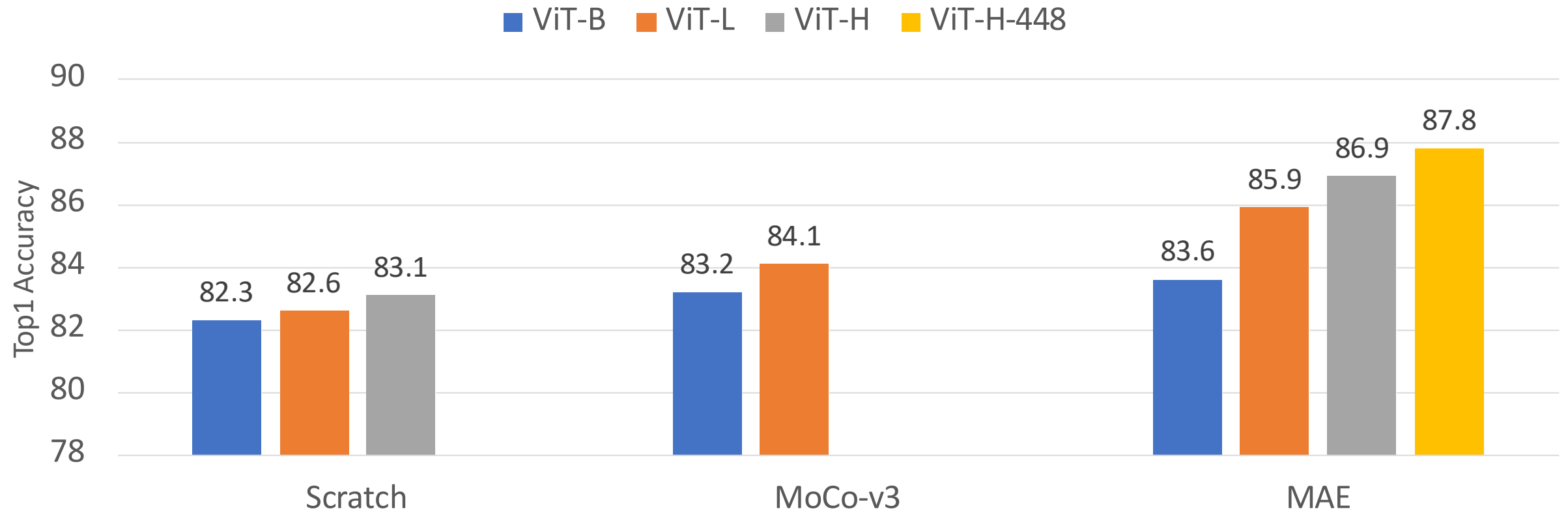
95% mask



MAE for Downstream Tasks: *Encoder Only*

- After MAE pre-training, just *throw away* the decoder
- Encoder is used for representations with *full-sequence* input

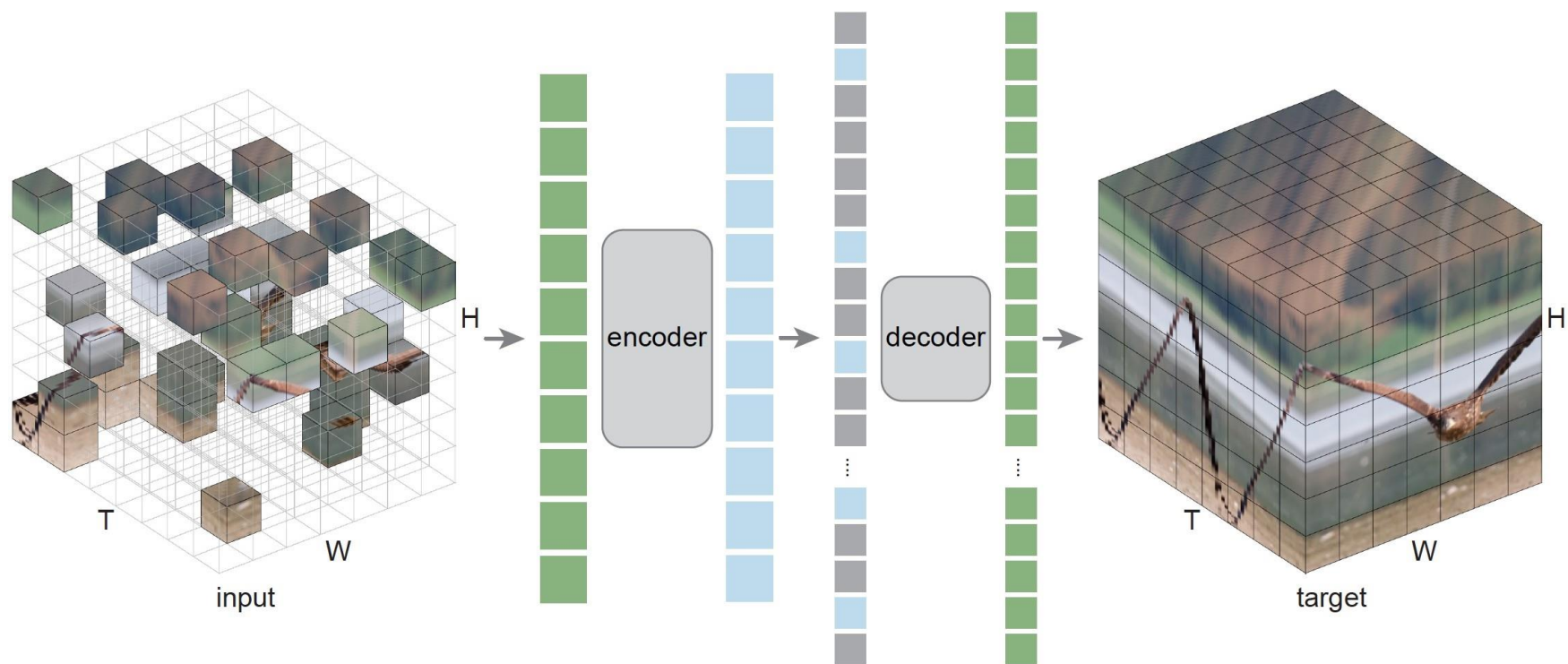




MAE Pretraining outperforms training from scratch, and allows scaling to larger ViT models

Reconstructive Approaches (Video Based)

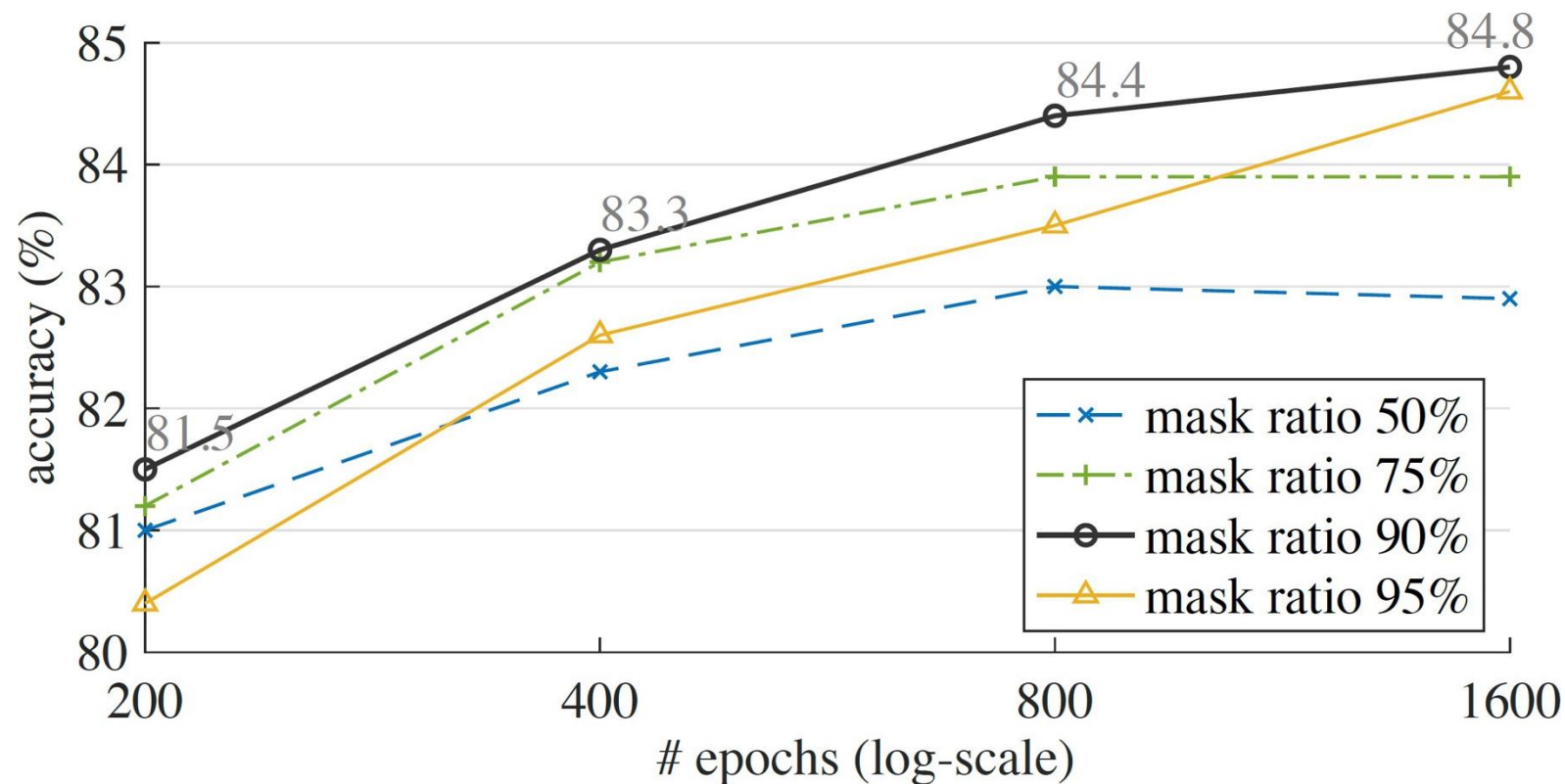
Masked Autoencoders as spatiotemporal learners



- Masking of random patches in spacetime
- Encoder operates on the set of visible patches
- A small decoder on encoded patches and mask tokens reconstruct input
- Except for patch and positional embeddings, no inductive bias

Masking ratio can be extremely high

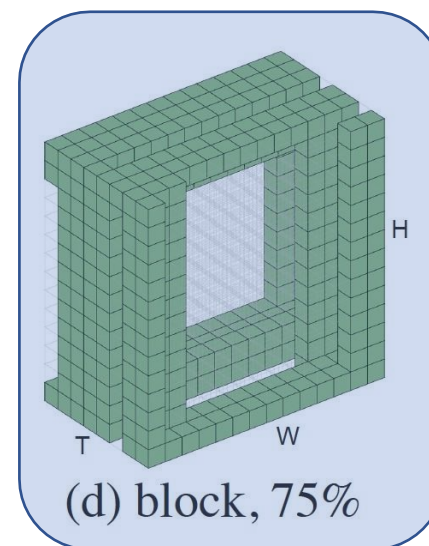
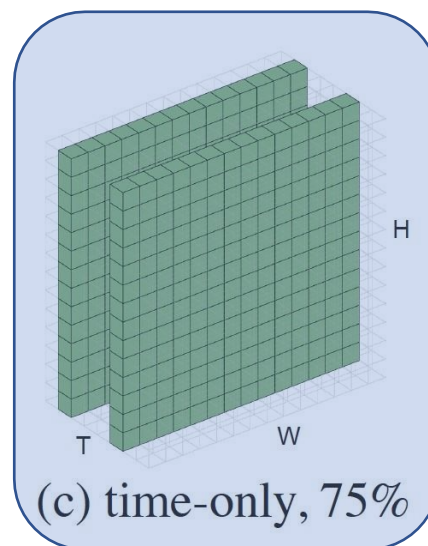
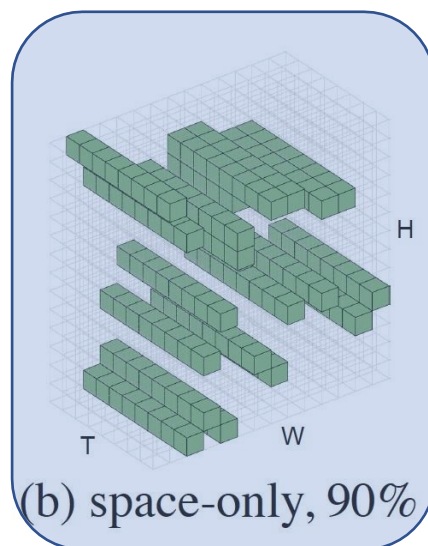
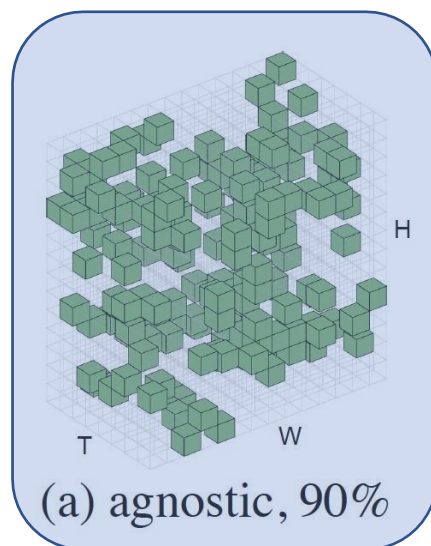
- Task:
Kinetics-400 (K400)
video classification
- Metric: accuracy (acc.)
- Model: ViT-L
- Pre-train: 200-1600 epochs
- Fine-tune: 100 epochs
- Training from scratch: 71.4%



- For image classification, 75% is the optimal value, but for video 90% is

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ct considerably better

Masking can be agnostic in spacetime



case	ratio	acc.
agnostic	90	84.4
space-only	90	83.5
time-only	75	79.1
block	75	83.2

MAE visualizations

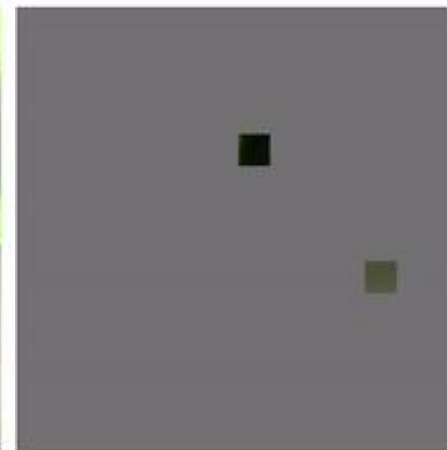
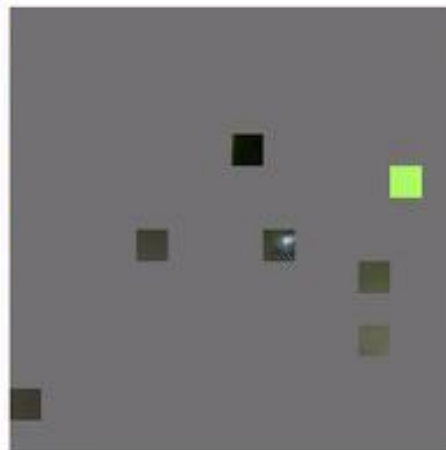
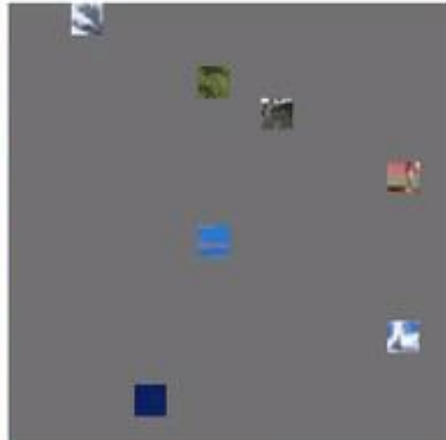
original

input 95% masked

output 95%

input 98% masked

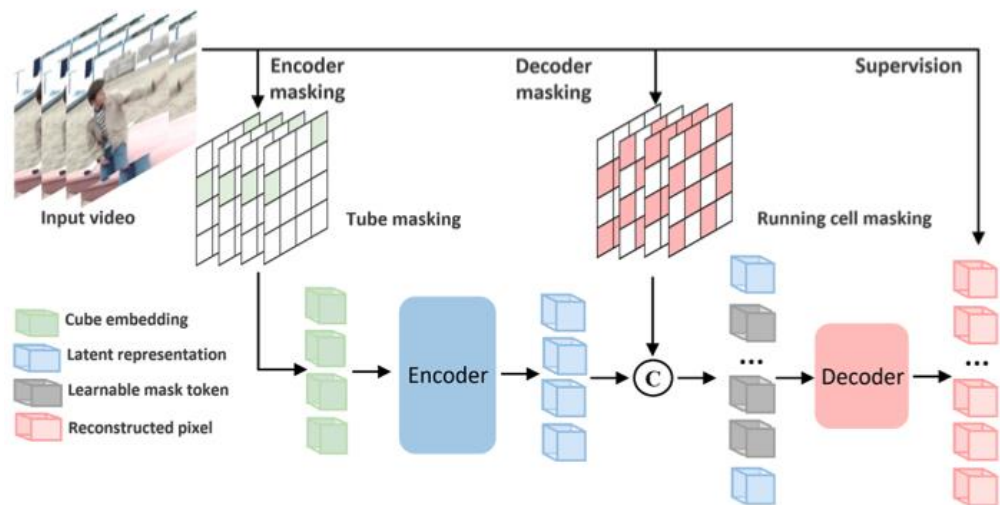
output 98%



VideoMAE v2

VideoMAE V2: Scaling Video Masked Autoencoders with Dual Masking

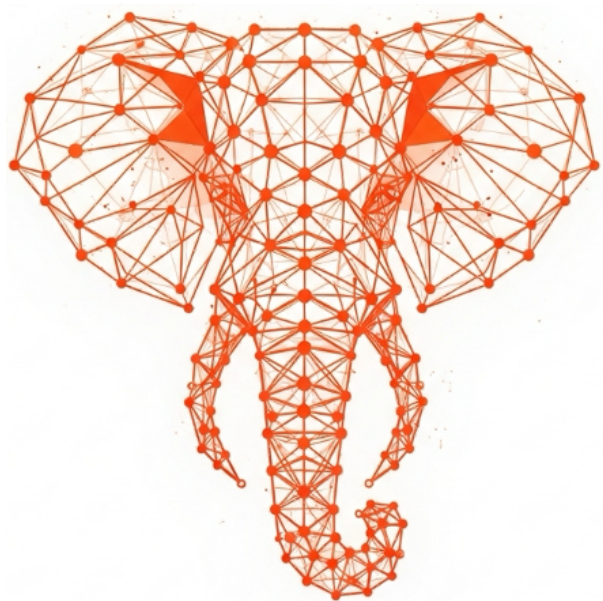
Limin Wang^{1,2,*} Bingkun Huang^{1,2,*} Zhiyu Zhao^{1,2} Zhan Tong¹
Yinan He² Yi Wang² Yali Wang^{3,2} Yu Qiao^{2,3}
¹ State Key Laboratory for Novel Software Technology, Nanjing University, China
² Shanghai AI Lab, China ³ Shenzhen Institute of Advanced Technology, CAS, China



- Trains on more data
- Introduces dual masking – adding masking to the decoder – to reduce computation
- Refines training techniques
- Trains a ViT with 1Billion parameters
- Reported results on the left are on Kinetics-400

method	pre-train data	data size	epoch	ViT-B	ViT-L	ViT-H	ViT-g
MAE-ST [28]	Kinetics400	0.24M	1600	81.3	84.8	85.1	-
MAE-ST [28]	IG-uncurated	1M	1600	-	84.4	-	-
VideoMAE V1 [90]	Kinetics400	0.24M	1600	81.5	85.2	86.6	-
VideoMAE V2	UnlabeledHybrid	1.35M	1200	81.5 (77.0)	85.4 (81.3)	86.9 (83.2)	87.2 (83.9)
$\Delta Acc.$ with V1	-	-	-	+ 0%	+ 0.2%	+ 0.3%	-

Wang, L., Huang, B., Zhao, Z., Tong, Z., He, Y., Wang, Y., ... & Qiao, Y. (2023). Videomae v2: Scaling video masked autoencoders with dual masking. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 14549-14560).



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ct **DEEP LEARNING**
ADVANCED MODELS AND METHODS

Deep Learning

Advanced Models & Methods

Prof. Antonino Furnari (antonino.furnari@unict.it)

Corso di Laurea Magistrale in Informatica

Dip. di Matematica e Informatica

Università di Catania

The End